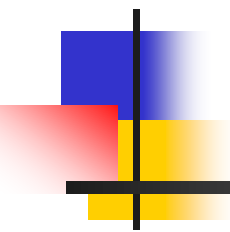


3D Computer Graphics



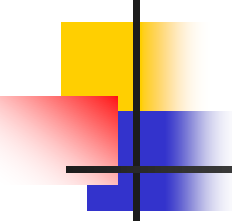
Rasterization II

Dan Biediger



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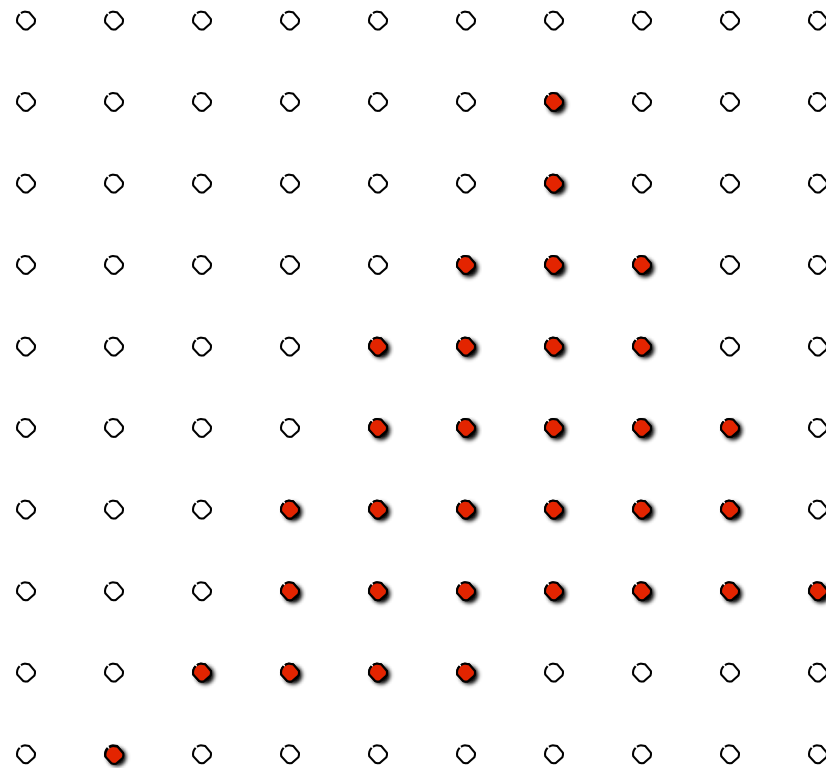


Today

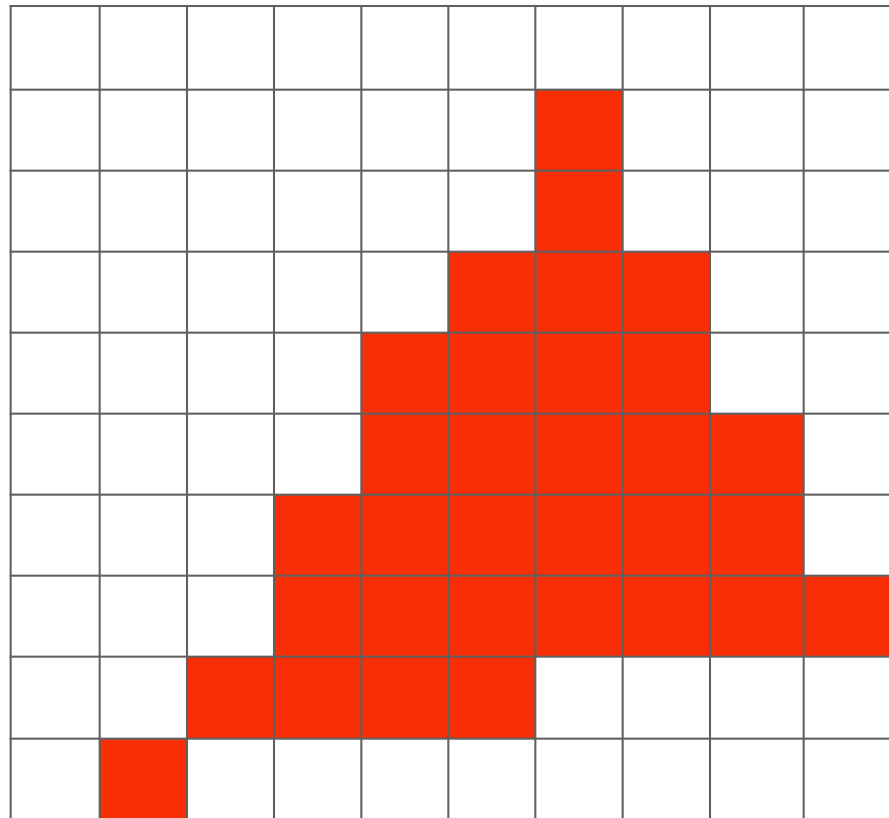
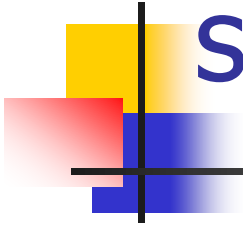
- Antialiasing
 - Sampling theory
 - Antialiasing in practice
- Visibility / occlusion
 - Z-buffering

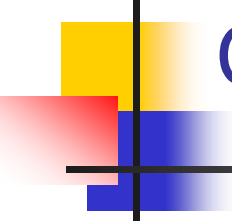


Recap: Testing in/out, at pixels' centers

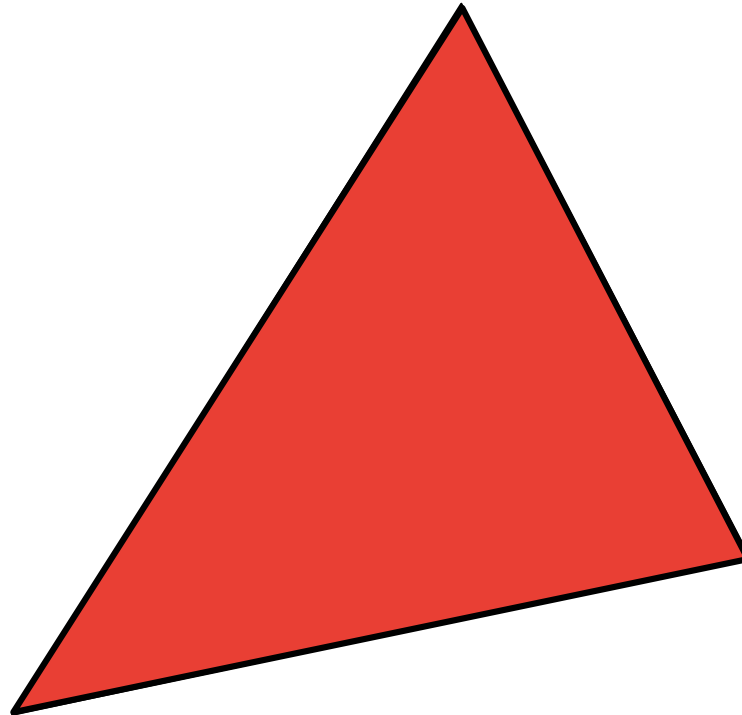


Pixels are uniformly-colored squares

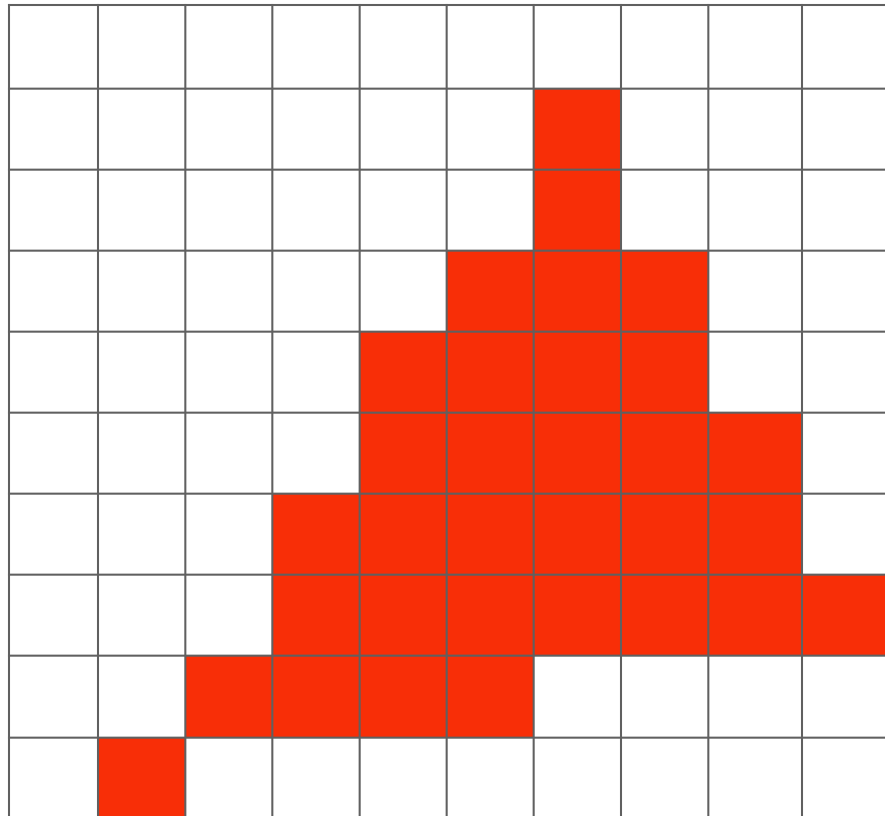




Compare: The Continuous Triangle Function



What's Wrong With This Picture?



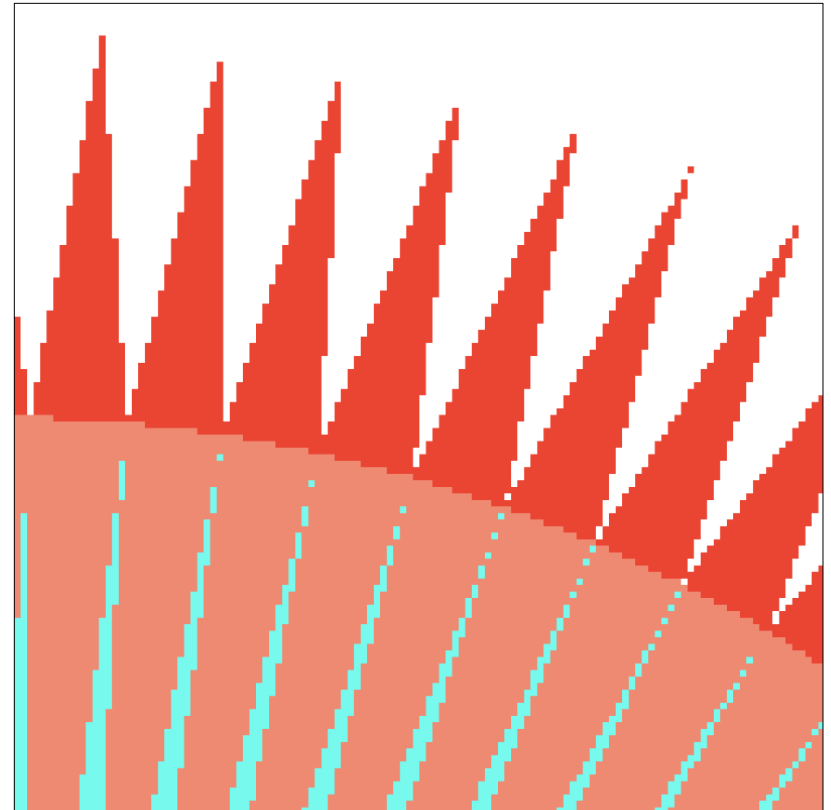
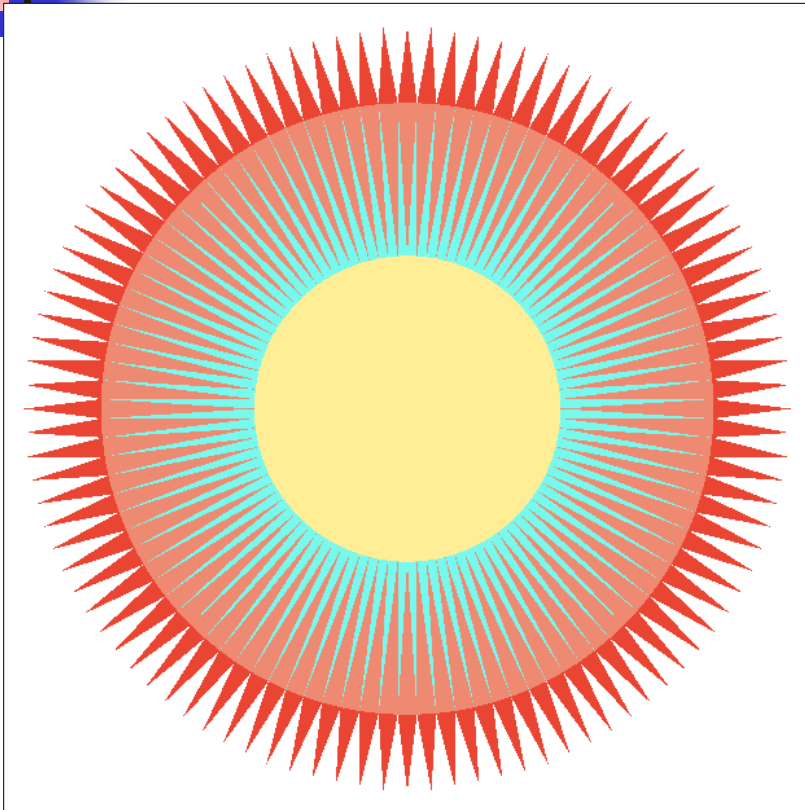
Jaggies!



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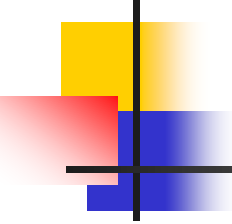
Aliasing



Is this the best we can do?

Slide courtesy of Prof. Ren Ng, UC Berkeley

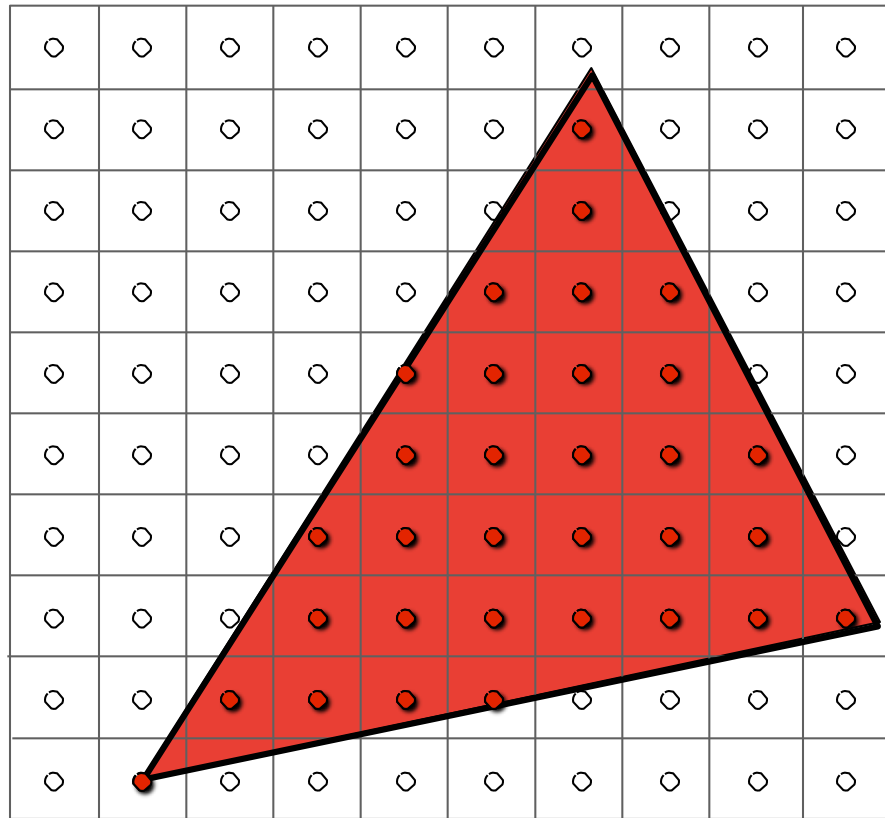




Sampling is Ubiquitous in Computer Graphics



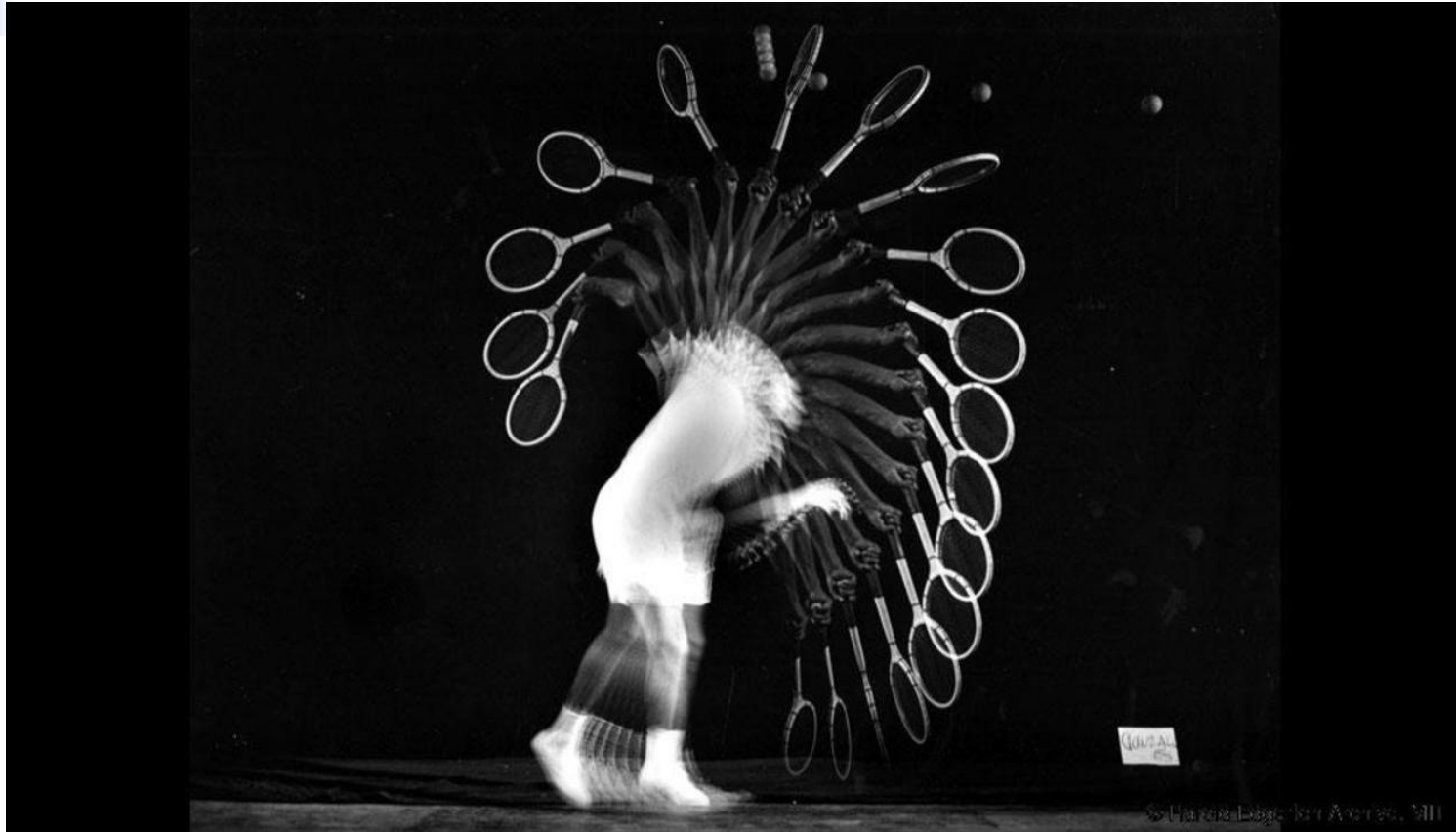
Rasterization = Sample 2D Positions



Photograph = Sample Image Sensor Plane



Video = Sample Time

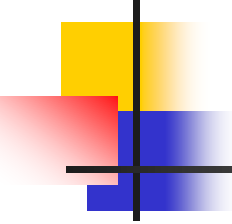


Harold Edgerton Archive, MIT



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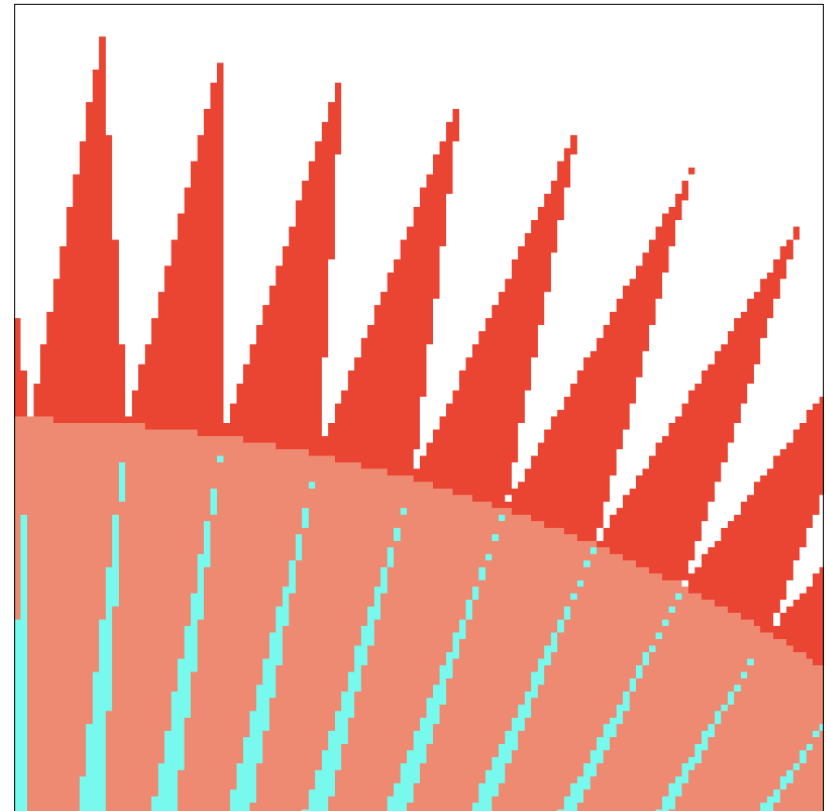
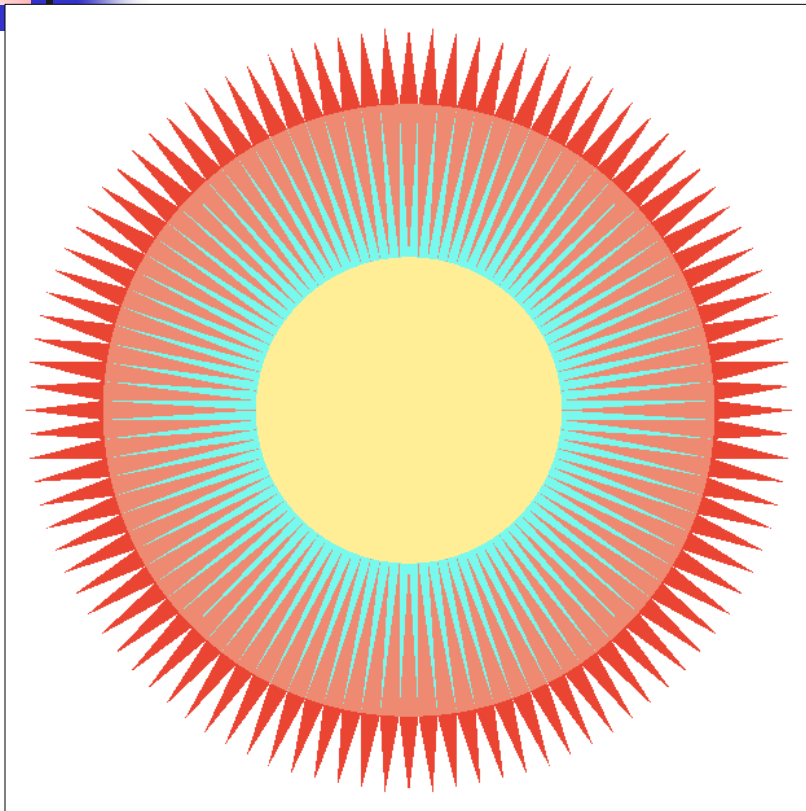
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Sampling **Artifacts** (Errors / Mistakes / Inaccuracies) in Computer Graphics



Jaggies (Staircase Pattern)



This is also an example of “aliasing” – a sampling error



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Moiré Patterns in Imaging



lystit.com

Skip odd rows and columns



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Wagon Wheel Illusion (False Motion)



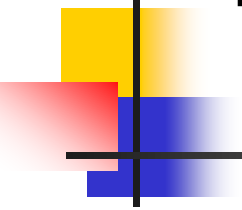
https://www.youtube.com/watch?v=QOwzkND_ooU



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Sampling Artifacts in Computer Graphics



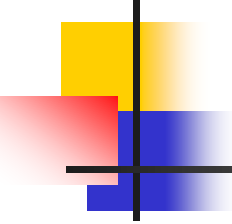
Artifacts due to sampling - “Aliasing”

- Jaggies – sampling in space
- Moire – undersampling images
- Wagon wheel effect – sampling in time
- [Many more] ...

Behind the Aliasing Artifacts

- Signals are **changing too fast** (high frequency), but **sampled too slowly**

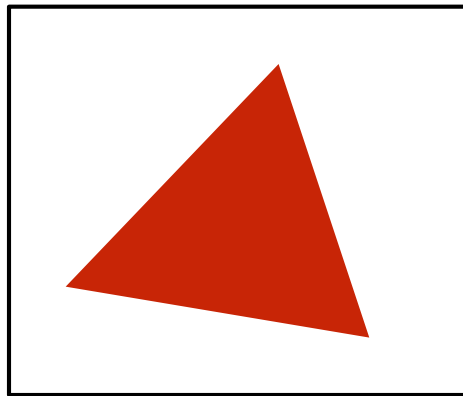




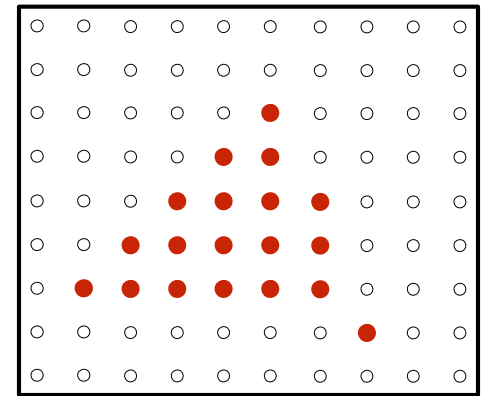
Antialiasing Idea: Blurring (Pre-Filtering) Before Sampling



Rasterization: Point Sampling in Space



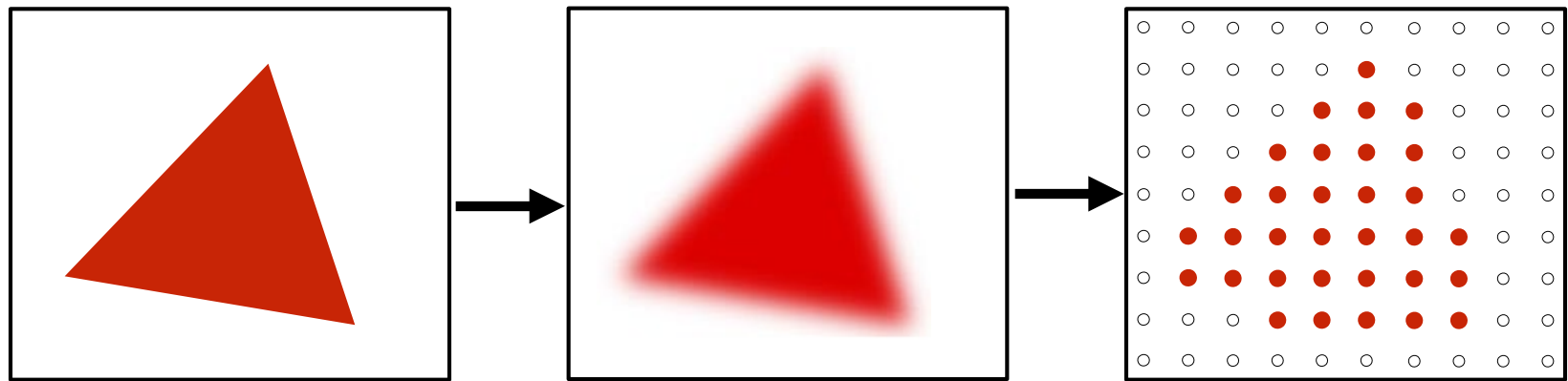
Sample



Note jaggies in rasterized triangle where pixel values are **pure red or white**



Rasterization: Antialiased Sampling



Pre-Filter

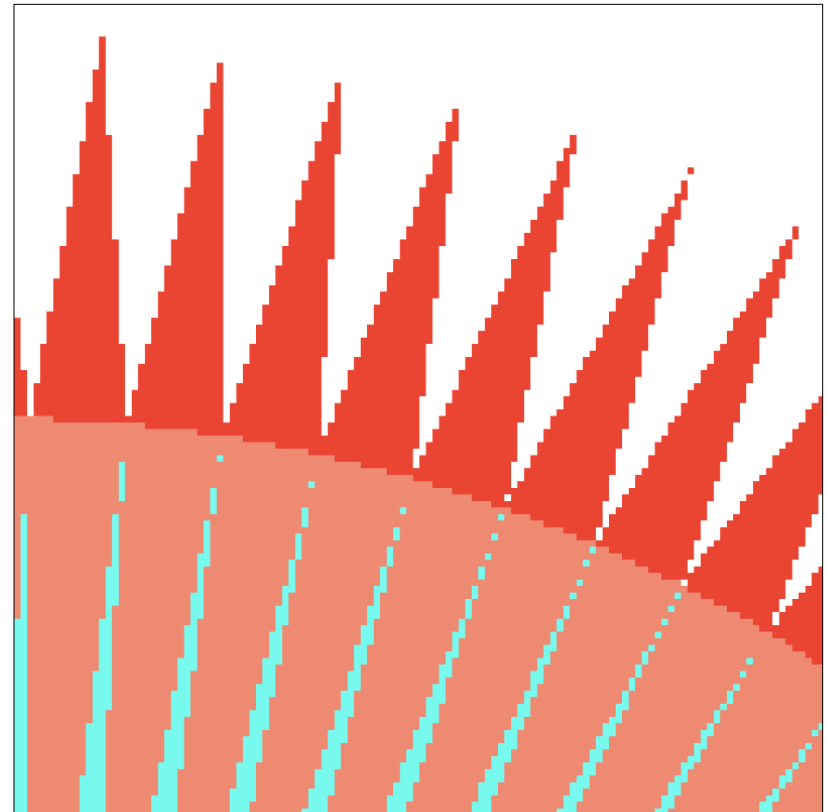
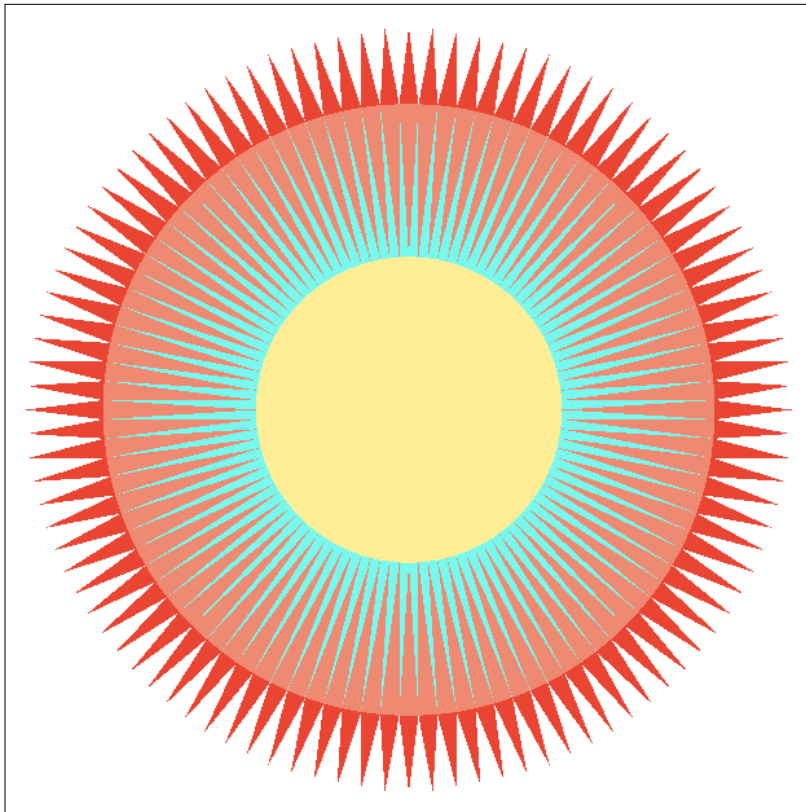
(remove frequencies above Nyquist)

Sample

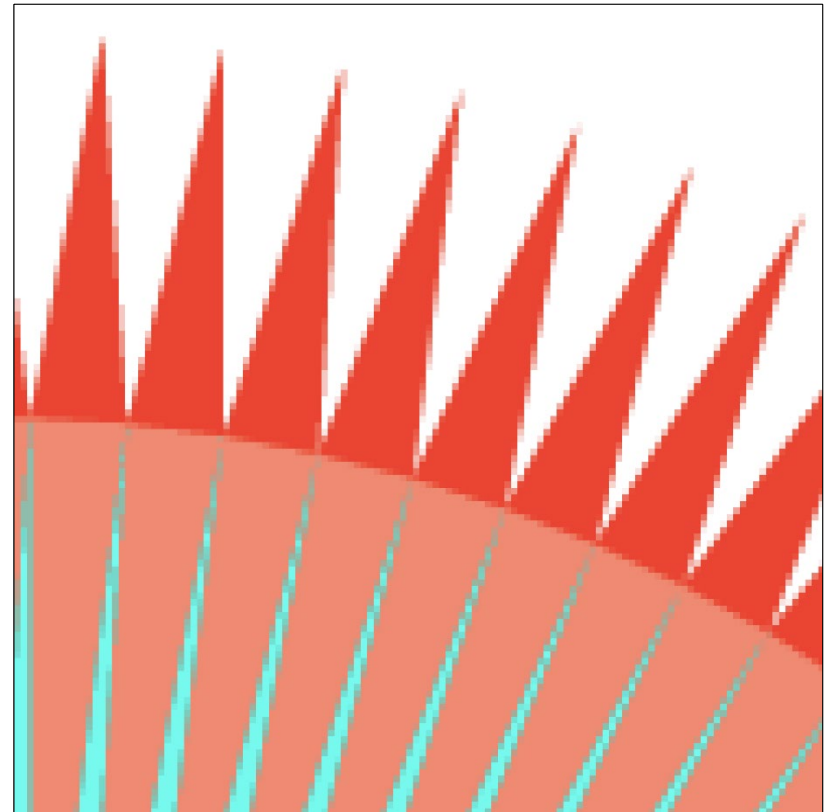
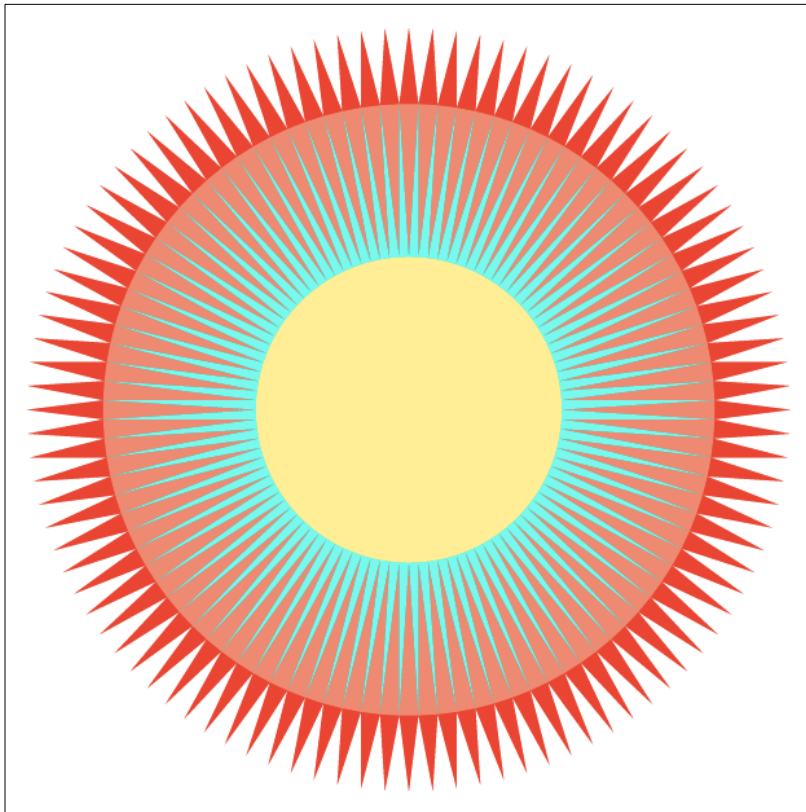
Note antialiased edges in rasterized triangle where pixel values take intermediate values



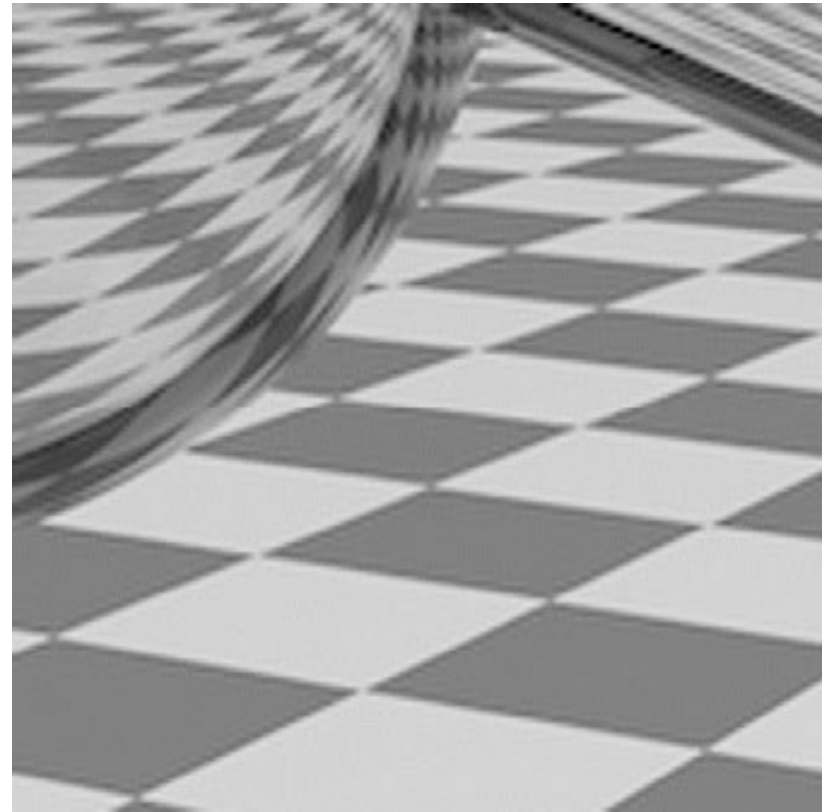
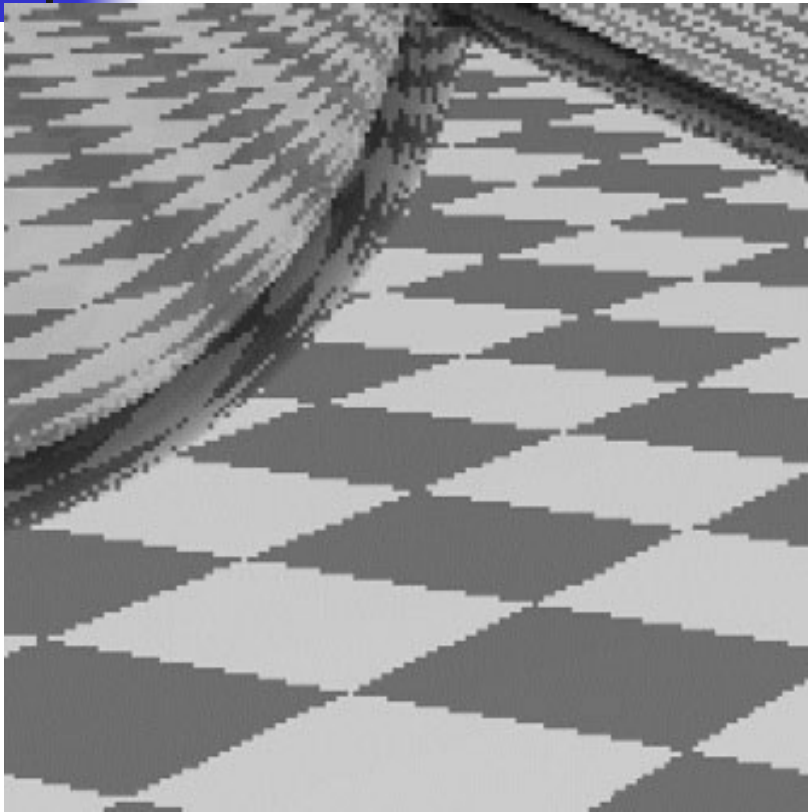
Point Sampling



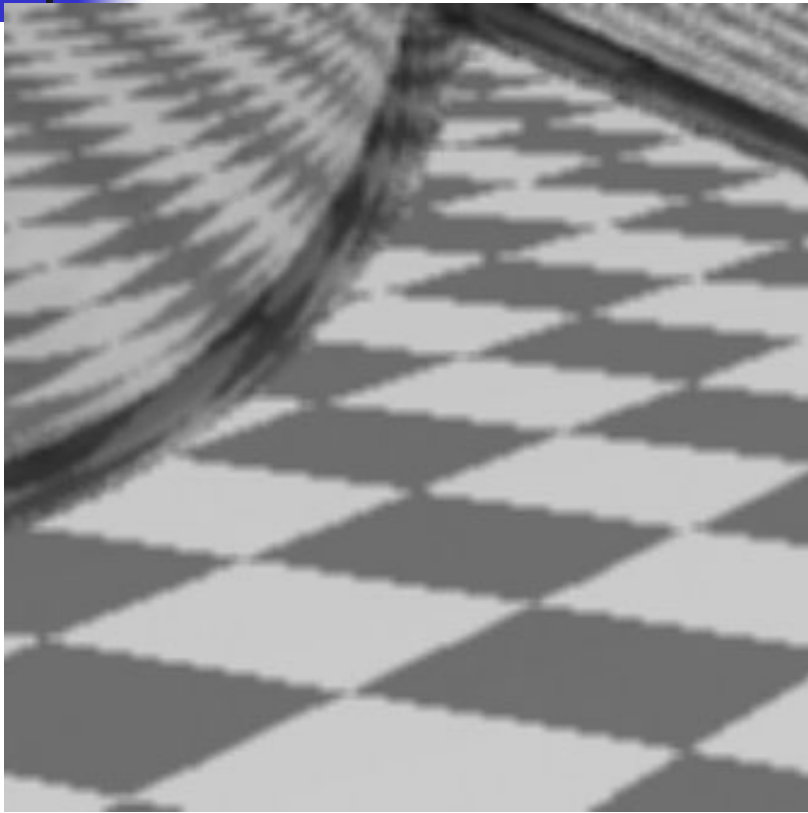
Antialiasing



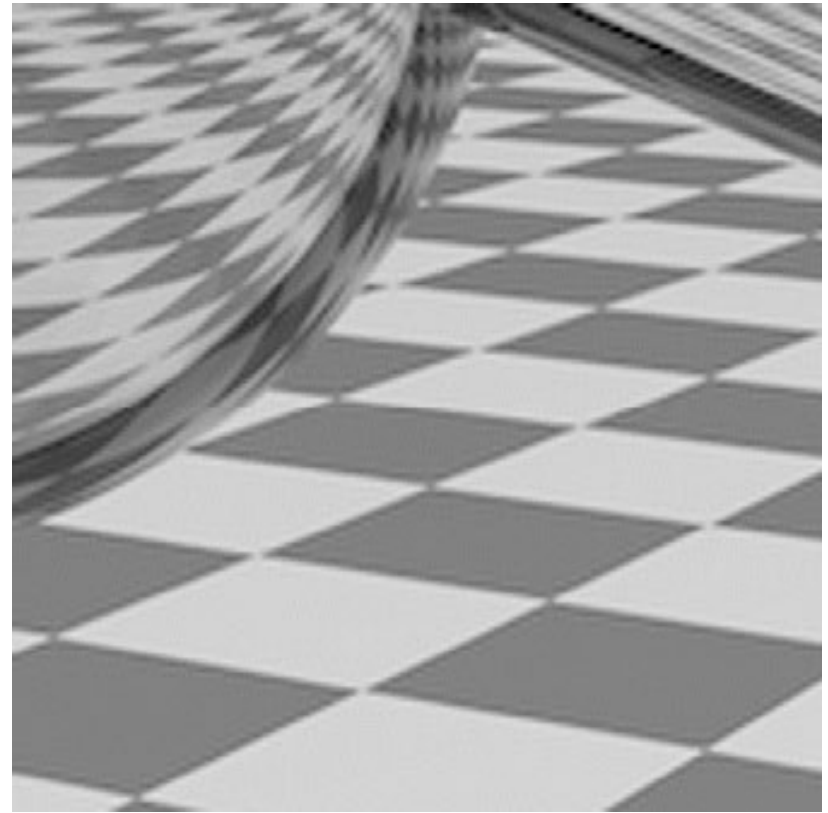
Point Sampling vs Antialiasing



Antialiasing vs Blurred Aliasing



(Sample then filter, WRONG!)

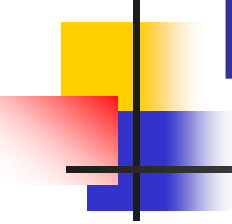


(Filter then sample)



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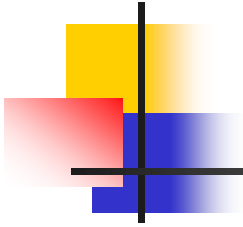
But why?

1. Why under-sampling introduces aliasing?
2. Why pre-filtering then sampling can do antialiasing?

Let's dig into fundamental reasons

And look at how to implement antialiased rasterization

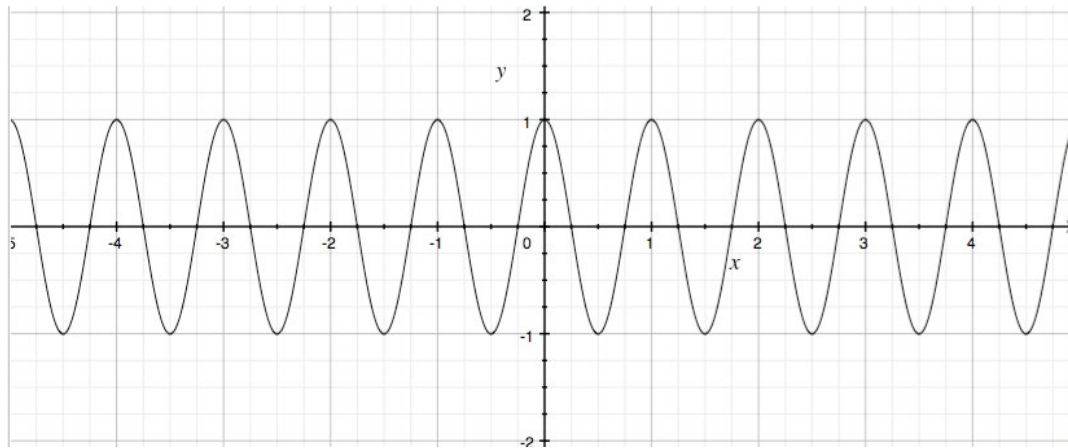




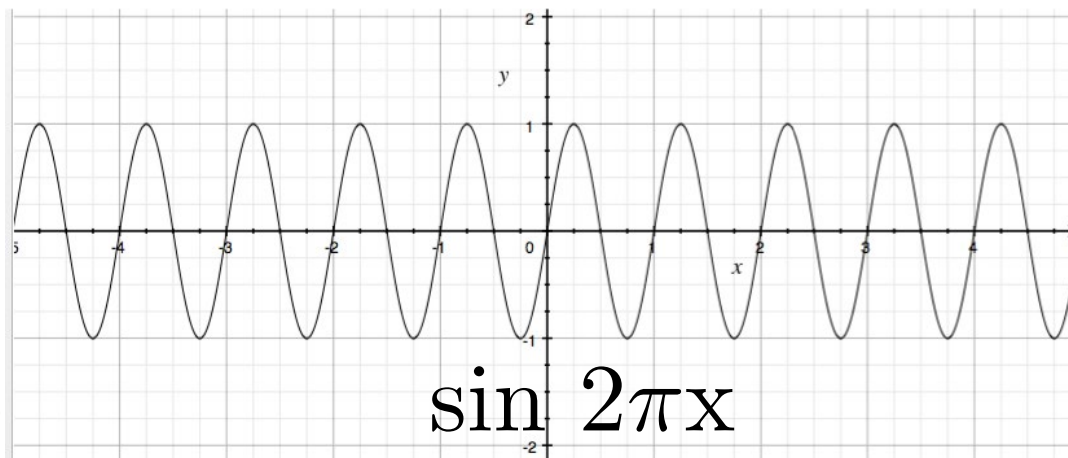
Frequency Domain



Sines and Cosines



$$\cos 2\pi x$$



$$\sin 2\pi x$$

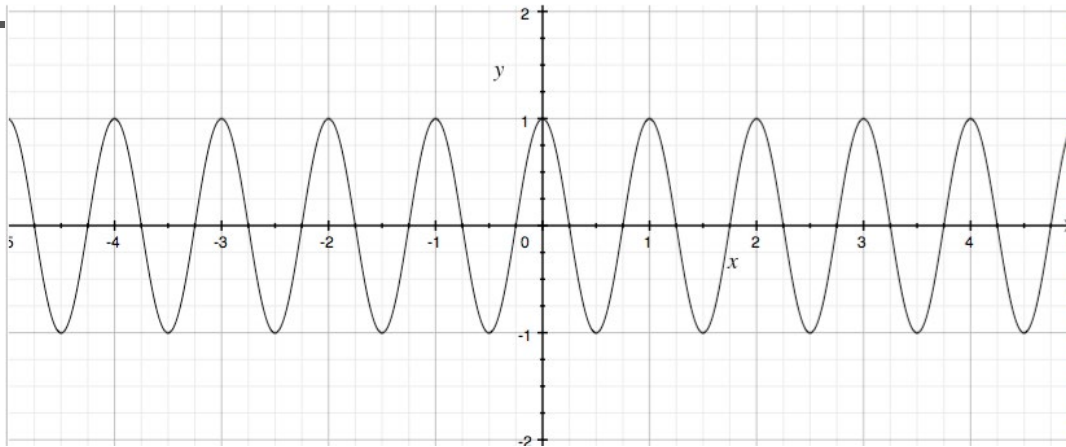


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Frequencies

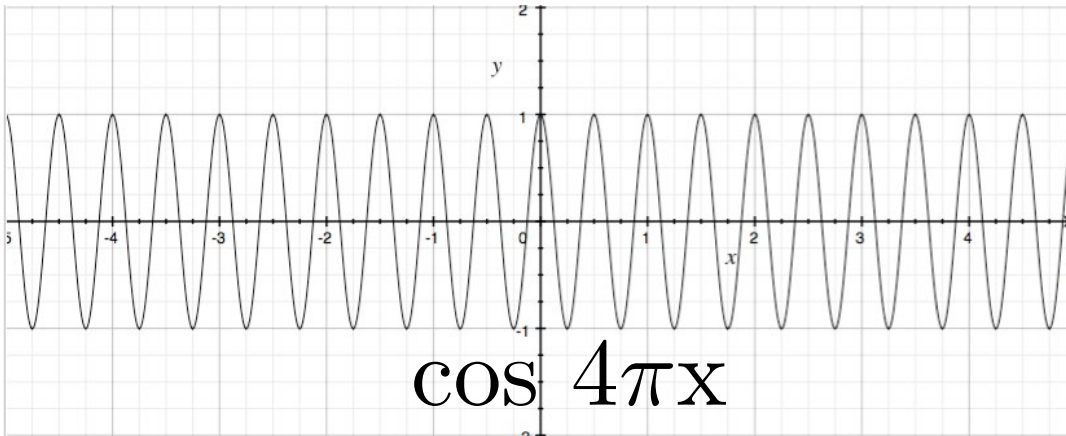
$$\cos 2\pi f x$$



$$f = 1$$

$$f = \frac{1}{T}$$

$$\cos 2\pi x$$



$$f = 2$$

$$\cos 4\pi x$$

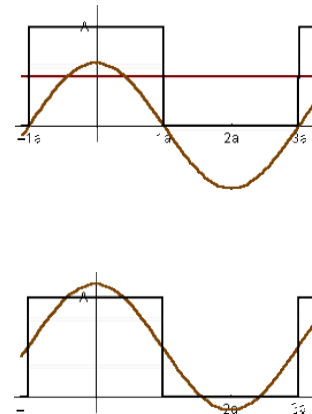


Fourier Transform

Represent a function as a
weighted sum of sines and
cosines



Joseph Fourier 1768 - 1830



$$f(x) = \frac{A}{2} + \frac{2A \cos(t\omega)}{\pi}$$

+ . . .

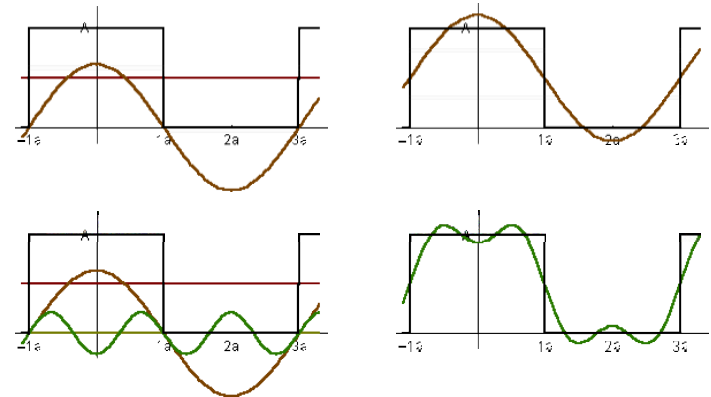


Fourier Transform

Represent a function as a weighted sum of sines and cosines



Joseph Fourier 1768 - 1830



$$f(x) = \frac{A}{2} + \frac{2A \cos(t\omega)}{\pi} - \frac{2A \cos(3t\omega)}{3\pi}$$

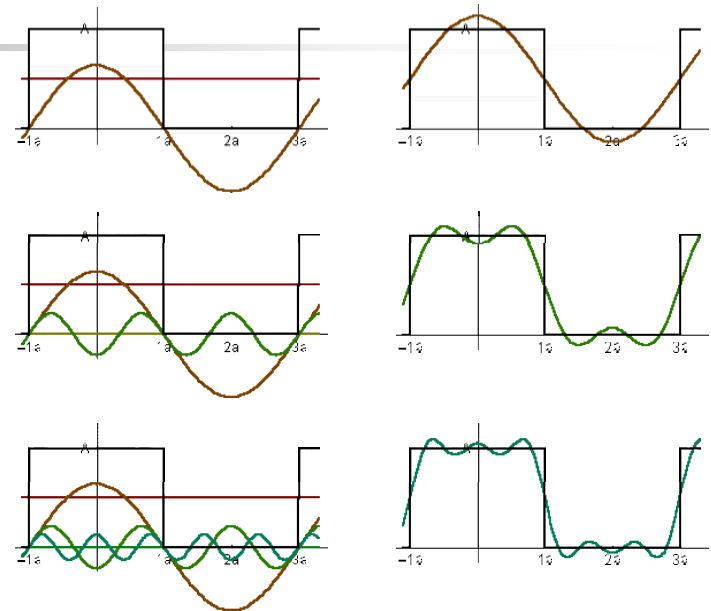


Fourier Transform

Represent a function as a weighted sum of sines and cosines



Joseph Fourier 1768 - 1830



$$f(x) = \frac{A}{2} + \frac{2A \cos(t\omega)}{\pi} - \frac{2A \cos(3t\omega)}{3\pi} + \frac{2A \cos(5t\omega)}{5\pi}$$

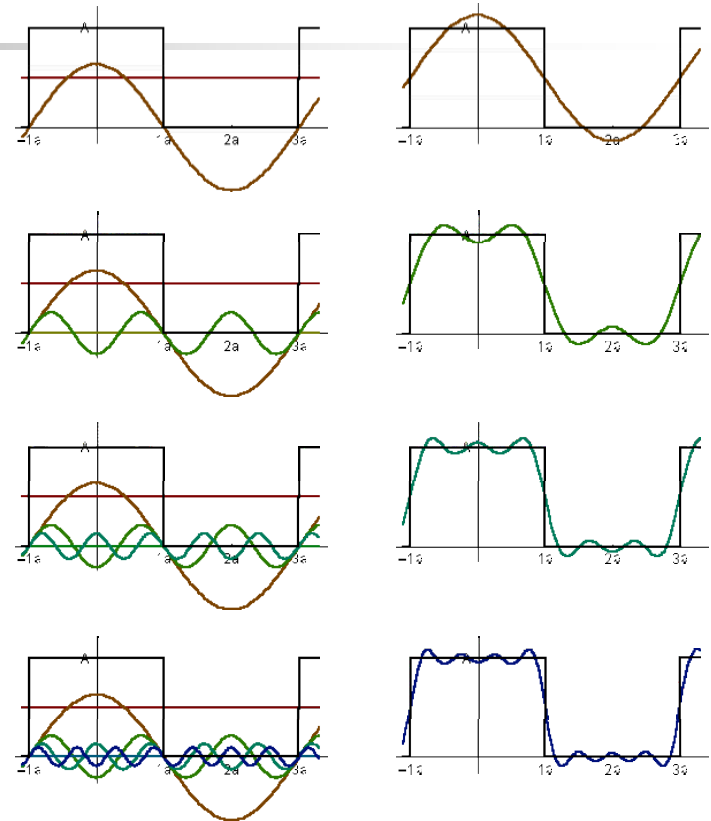


Fourier Transform

Represent a function as a weighted sum of sines and cosines



Joseph Fourier 1768 - 1830



$$f(x) = \frac{A}{2} + \frac{2A \cos(t\omega)}{\pi} - \frac{2A \cos(3t\omega)}{3\pi} + \frac{2A \cos(5t\omega)}{5\pi} - \frac{2A \cos(7t\omega)}{7\pi}$$

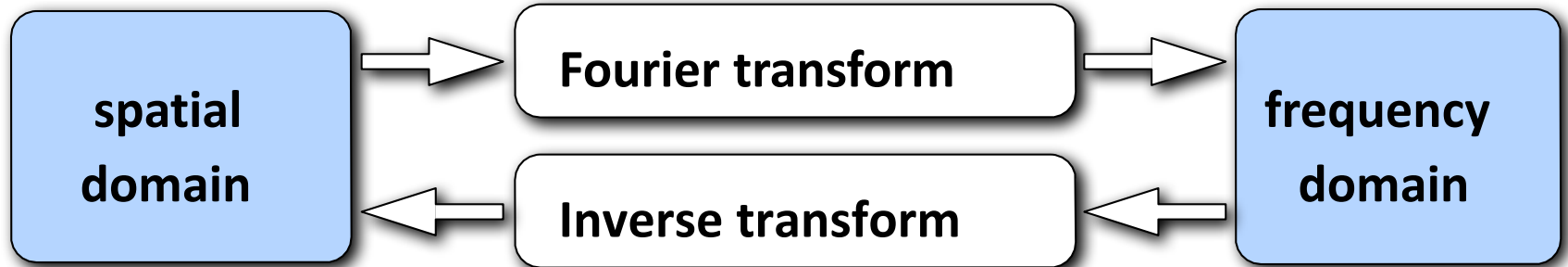


Fourier Transform Decomposes A Signal Into Frequencies

$f(x)$

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \omega x} dx$$

$F(\omega)$

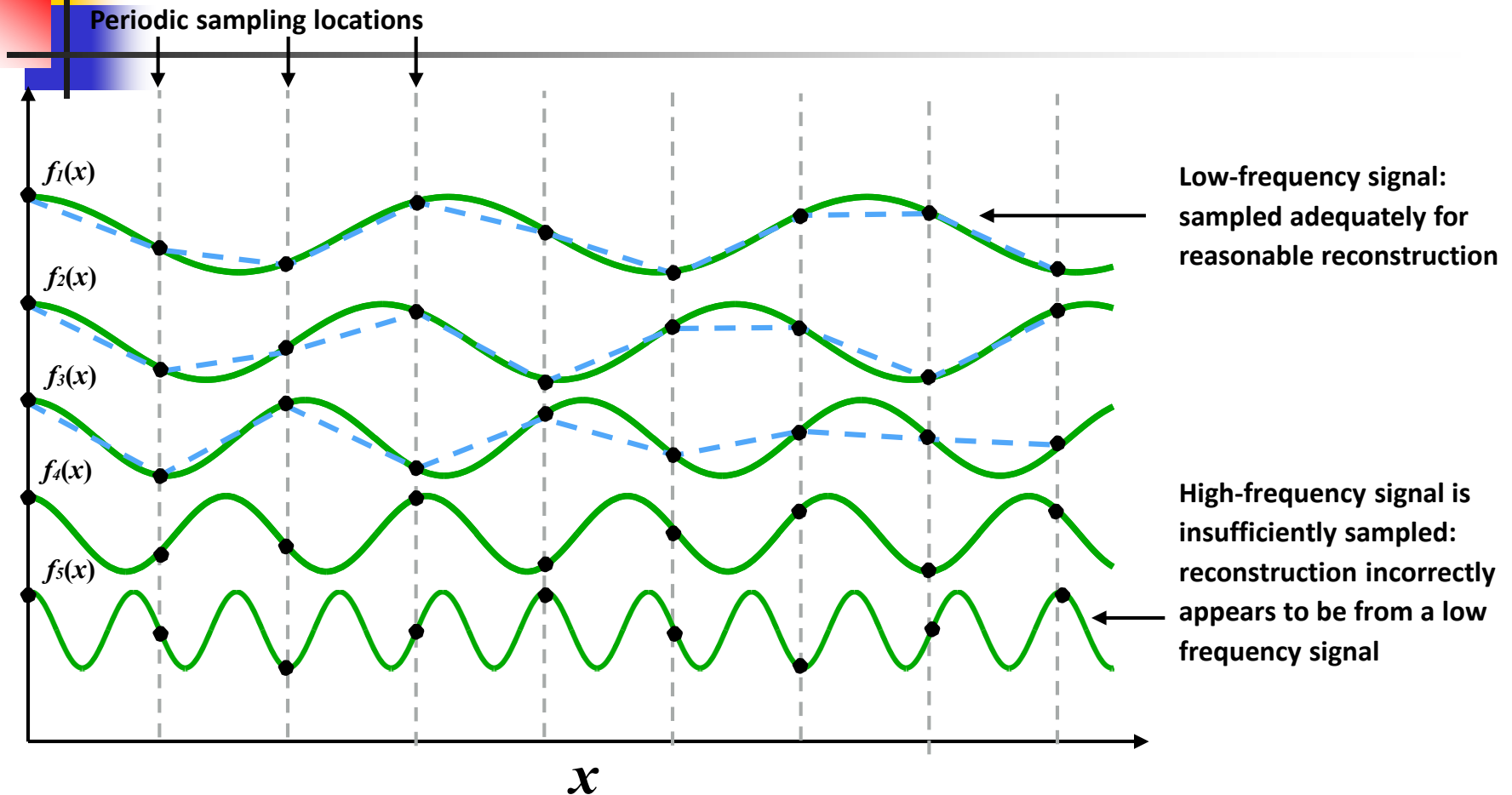


$$f(x) = \int_{-\infty}^{\infty} F(\omega) e^{2\pi i \omega x} d\omega$$

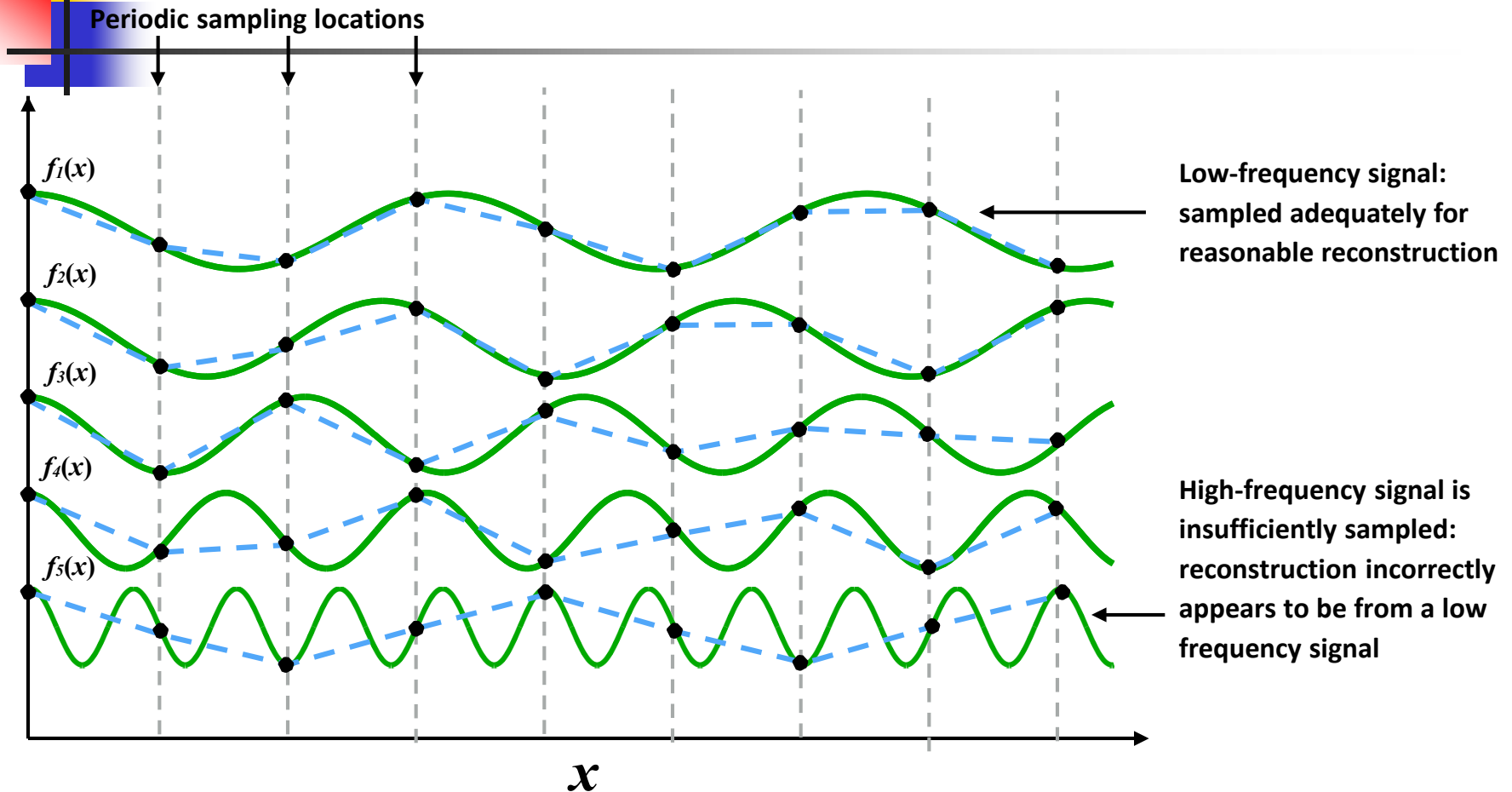
Recall $e^{ix} = \cos x + i \sin x$



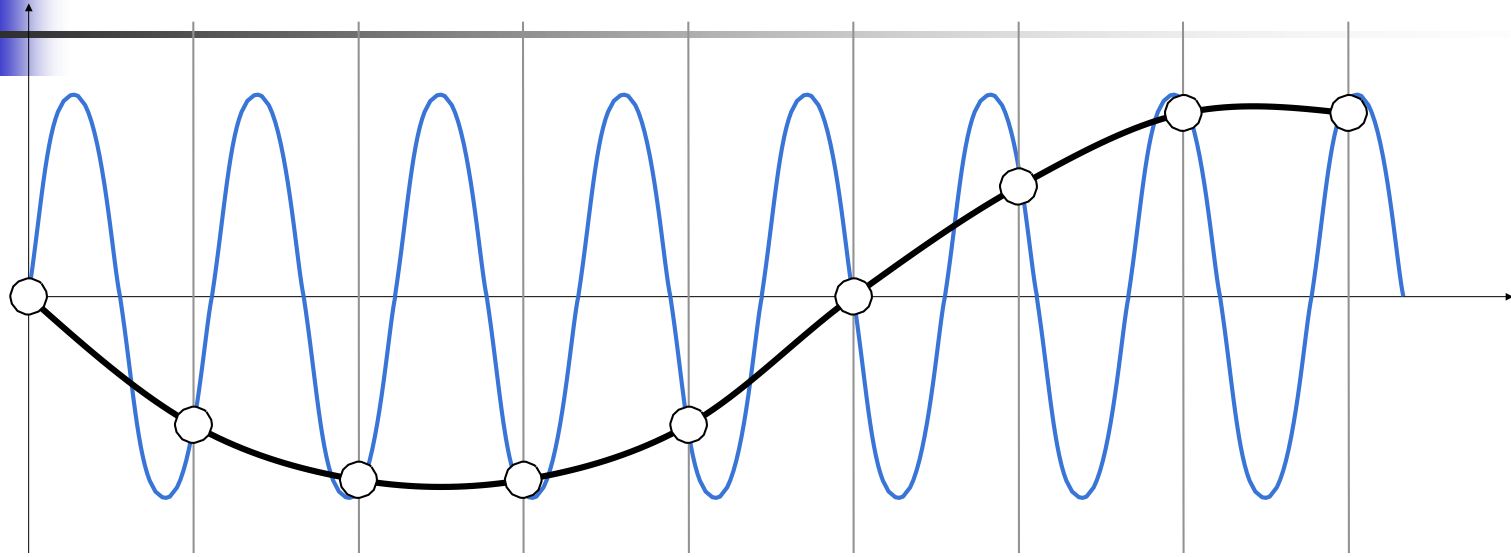
Higher Frequencies Need Faster Sampling



Higher Frequencies Need Faster Sampling



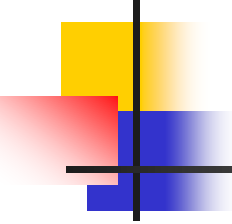
Undersampling Creates Frequency Aliases



High-frequency signal is insufficiently sampled: samples erroneously appear to be from a low-frequency signal

Two frequencies that are indistinguishable at a given sampling rate are called “**aliases**” (*misidentification of a signal frequency, introducing distortion or error*).

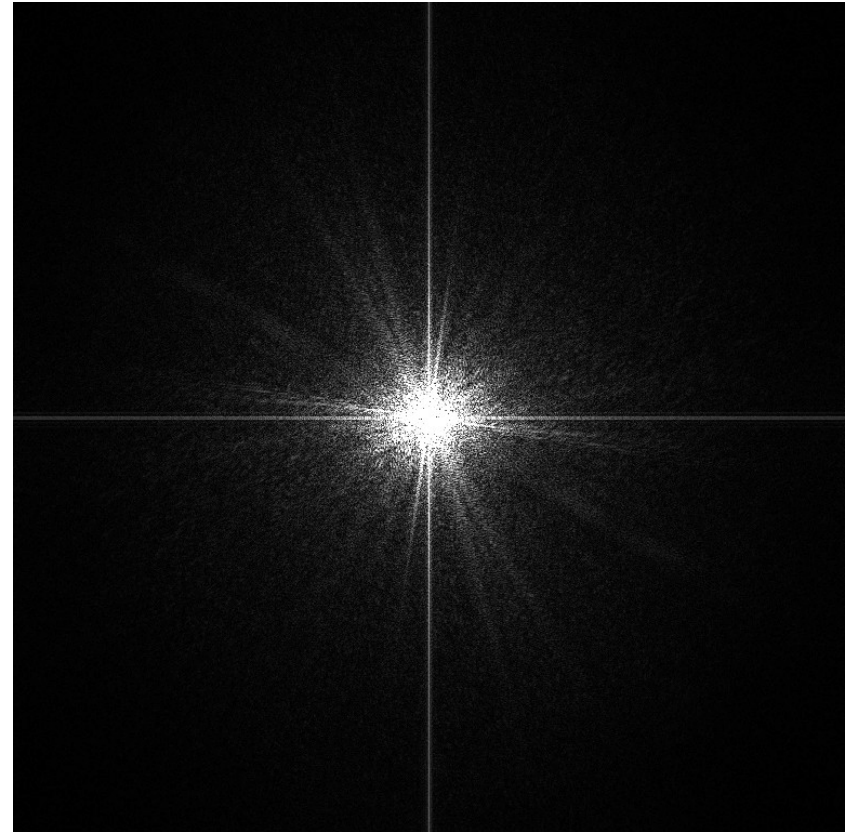




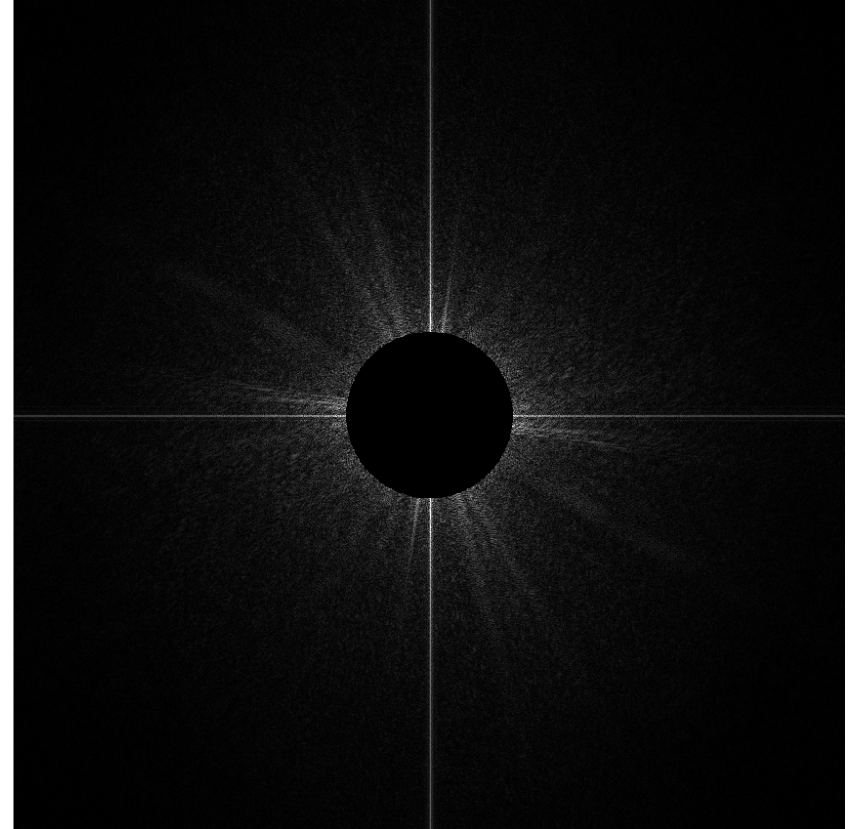
Filtering = Getting rid of
certain frequency contents



Visualizing Image Frequency Content



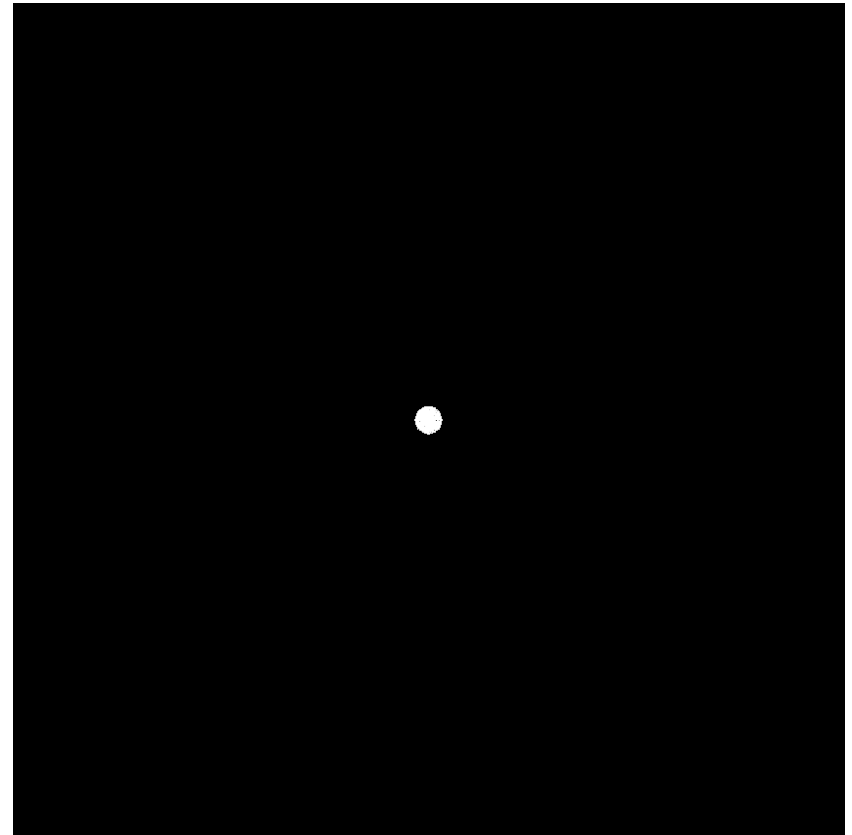
Filter Out Low Frequencies Only (Edges)



High-pass filter



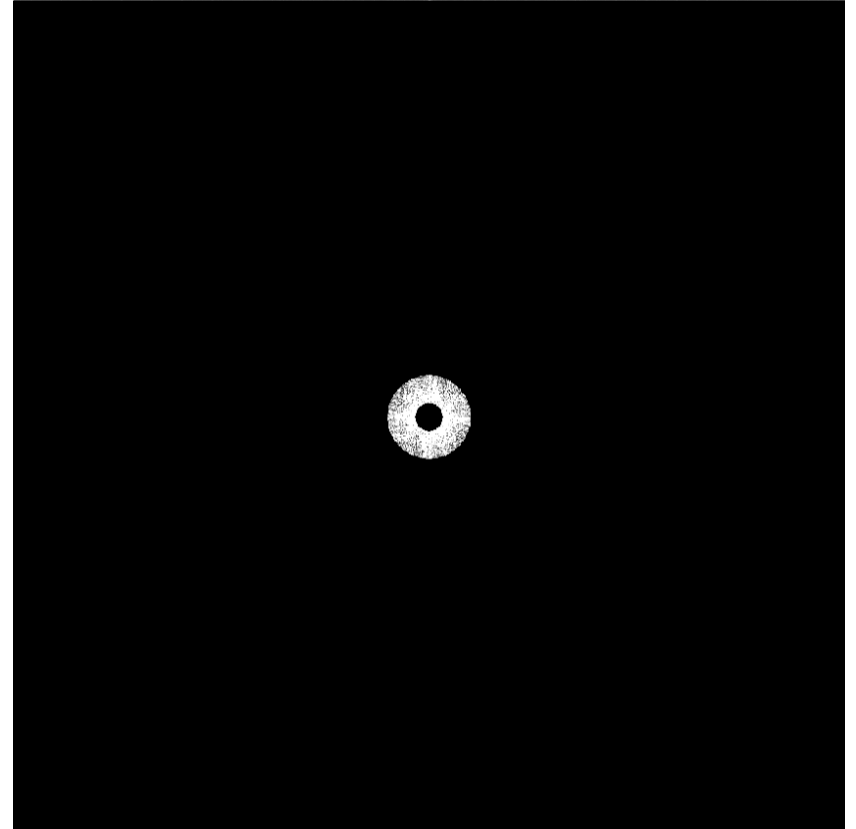
Filter Out High Frequencies (Blur)



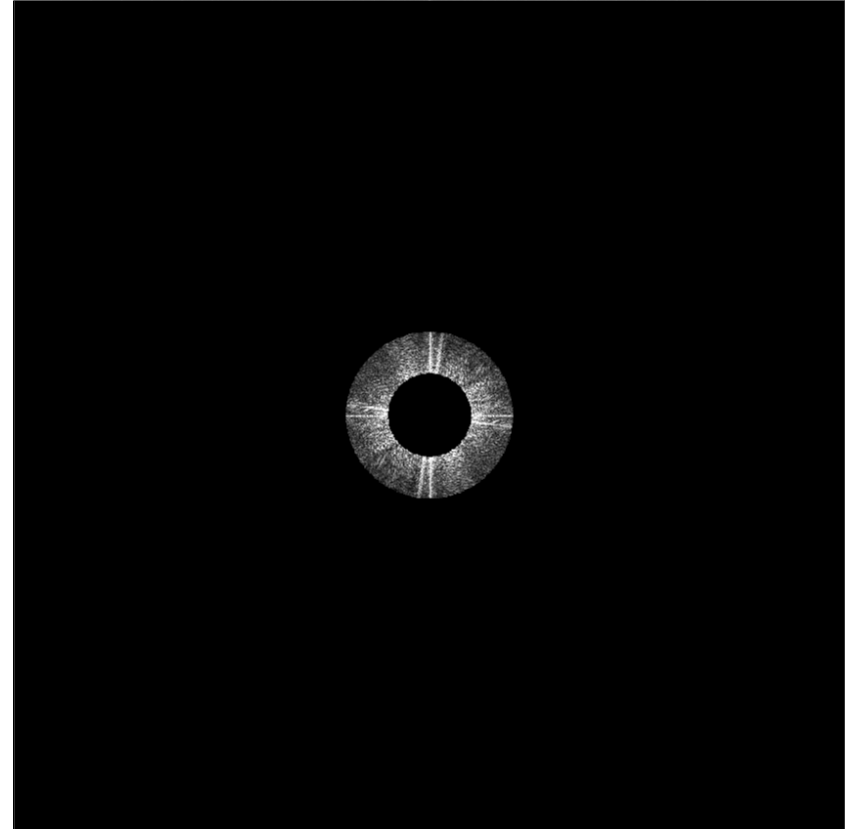
Low-pass filter

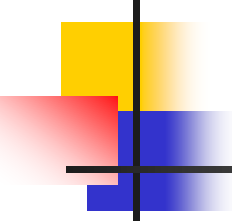


Filter Out Low and High Frequencies



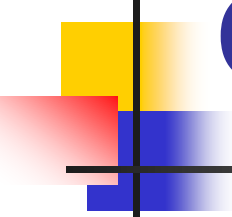
Filter Out Low and High Frequencies





Filtering = Convolution (= Averaging)





Convolution

Signal

1	3	5	3	7	1	3	8	6	4
---	---	---	---	---	---	---	---	---	---

Filter

$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
---------------	---------------	---------------

Point-wise local averaging in a “sliding window”



Convolution

Signal

1	3	5	3	7	1	3	8	6	4
---	---	---	---	---	---	---	---	---	---

Filter

1/4	1/2	1/4
-----	-----	-----

$$1 \times (1/4) + 3 \times (1/2) + 5 \times (1/4) = 3$$

Result

	3								
--	---	--	--	--	--	--	--	--	--



Convolution

Signal

1	3	5	3	7	1	3	8	6	4
---	---	---	---	---	---	---	---	---	---

Filter

1/4	1/2	1/4
-----	-----	-----

$$3 \times (1/4) + 5 \times (1/2) + 3 \times (1/4) = 4$$

Result

	3	4							
--	---	---	--	--	--	--	--	--	--





Convolution Theorem

Convolution in the spatial domain is **equivalent to multiplication in the frequency domain**, and vice versa

Option 1:

- Filter by convolution in the spatial domain

Option 2:

- Transform to frequency domain (Fourier transform)
- Multiply by Fourier transform of convolution kernel
- Transform back to spatial domain (inverse Fourier)



Convolution Theorem

Spatial
Domain



*

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

=



Fourier
Transform



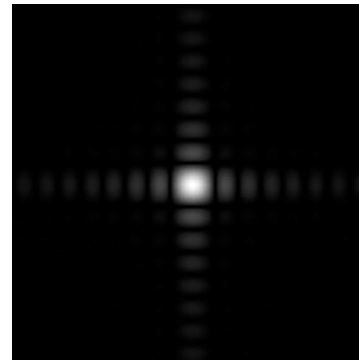
Inv. Fourier
Transform



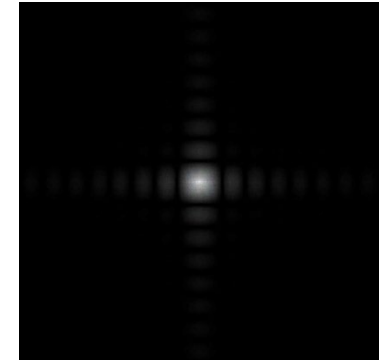
Frequency
Domain

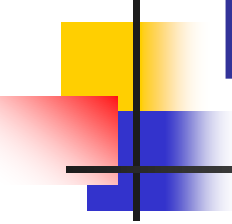


x



=





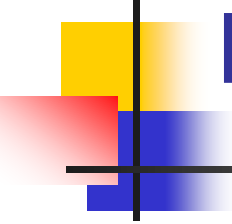
Box Filter

$$\frac{1}{9}$$

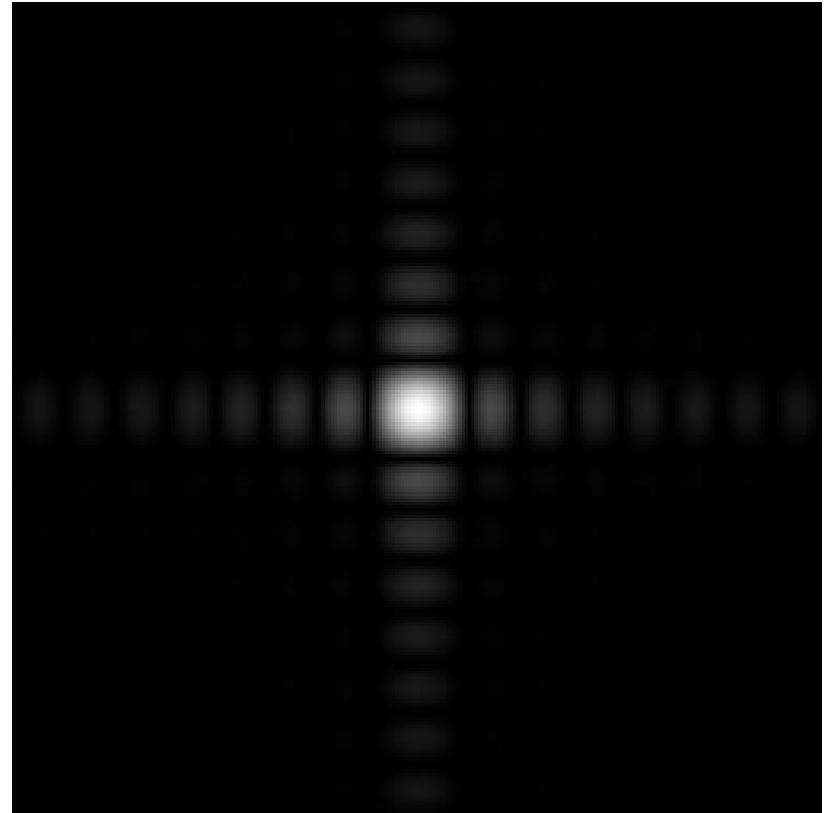
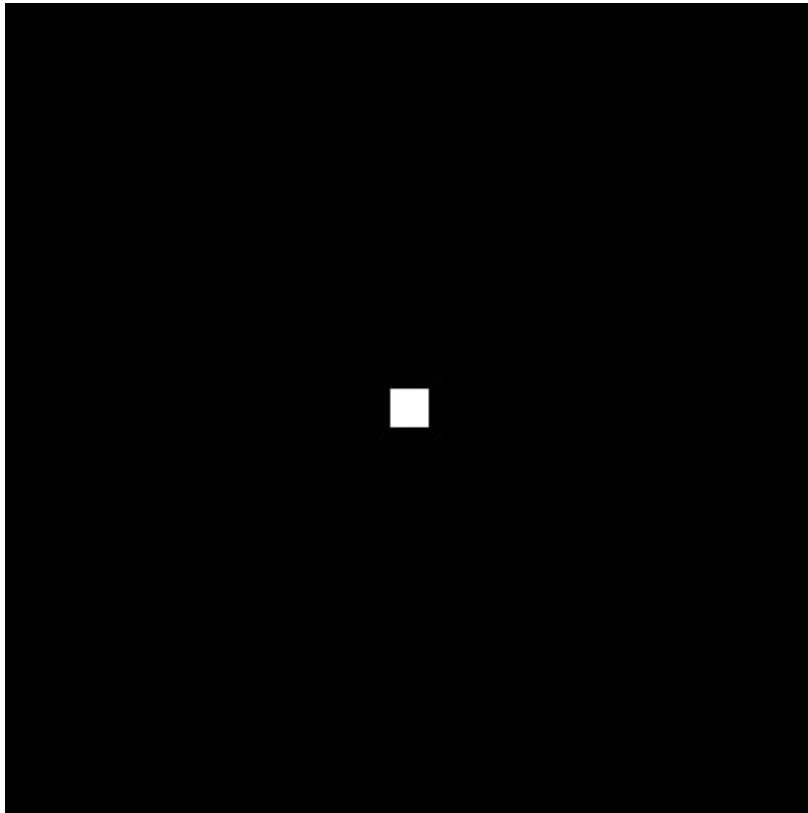
1	1	1
1	1	1
1	1	1

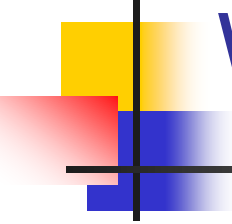
Example: 3x3 box filter



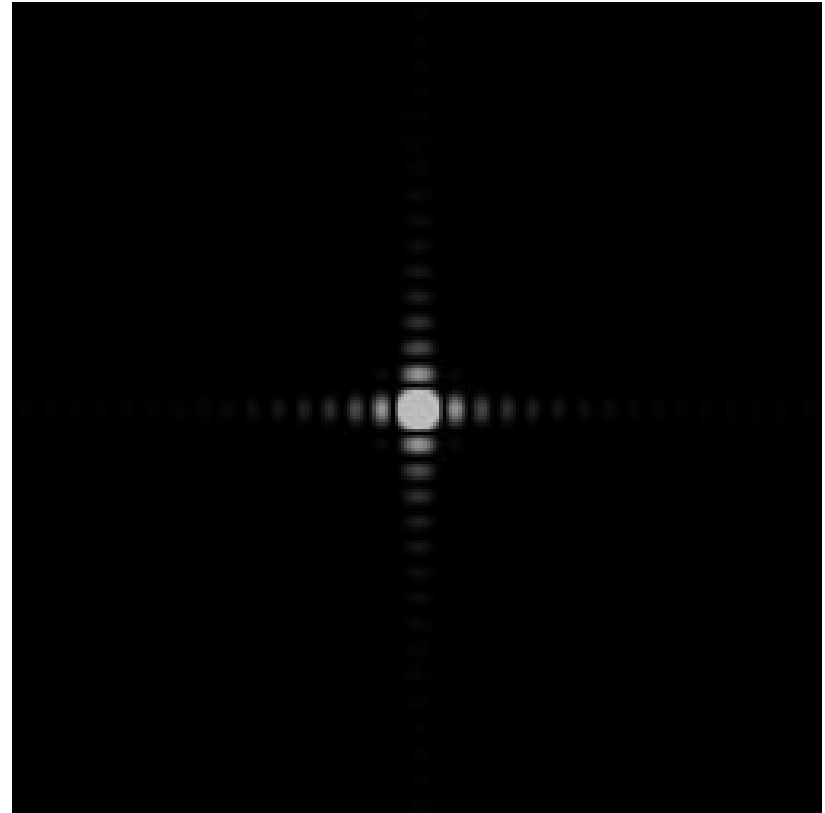
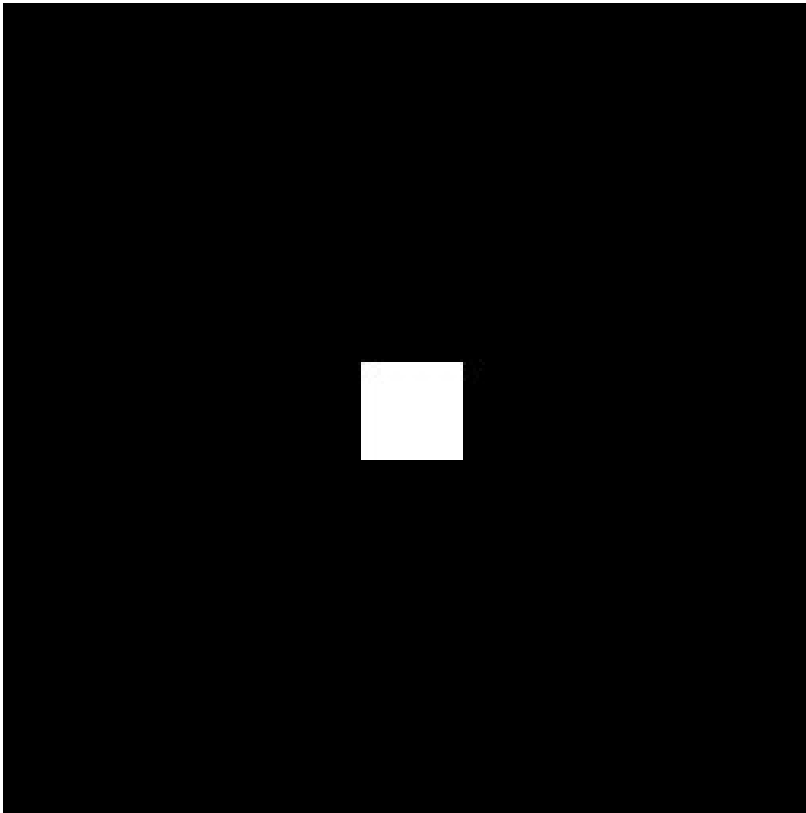


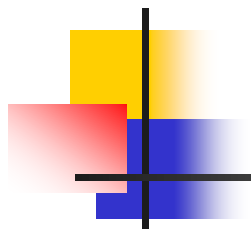
Box Function, “Low Pass” Filter





Wider Filter Kernel, Lower Frequencies





Antialiasing



How Can We Reduce Aliasing Error?

Option 1: Increase sampling rate

- Essentially increasing the distance between replicas in the Fourier domain
- Higher resolution displays, sensors, framebuffers...
- But: costly & may need very high resolution

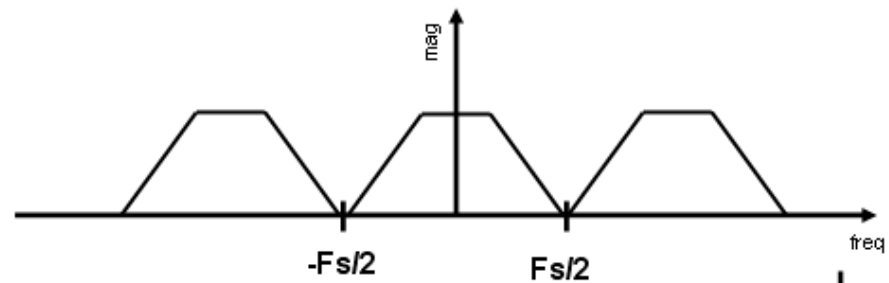
Option 2: Antialiasing

- Making Fourier contents “narrower” before repeating
- i.e. **Filtering out high frequencies before sampling**

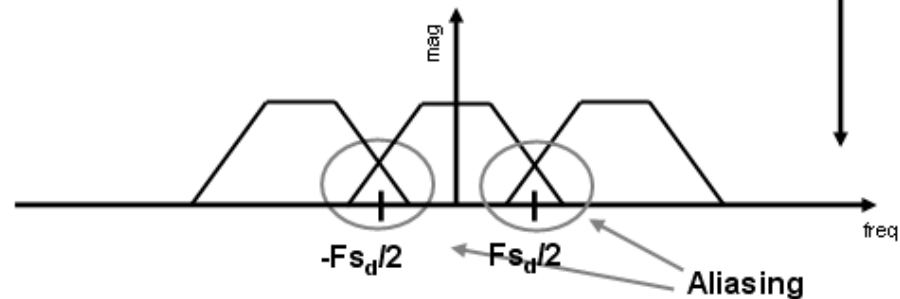


Aliasing = Mixed Frequency Contents

Dense sampling:

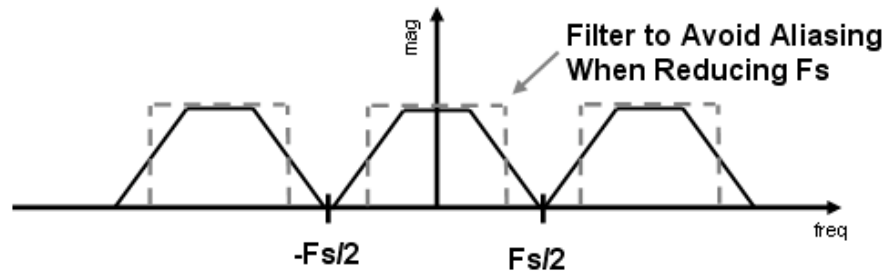


Sparse sampling:

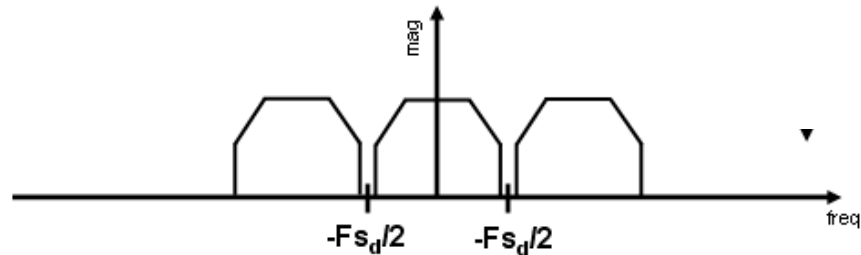


Antialiasing = Limiting, then repeating

Filtering



Then sparse sampling



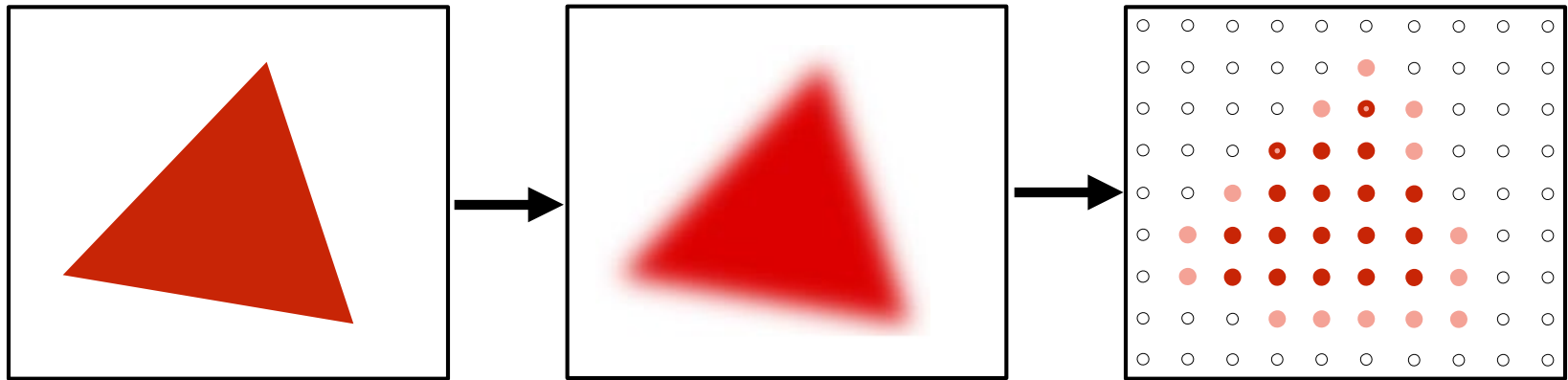
Regular Sampling



Note jaggies in rasterized triangle where pixel values are pure red or white



Antialiased Sampling



Pre-Filter

(remove frequencies above Nyquist)

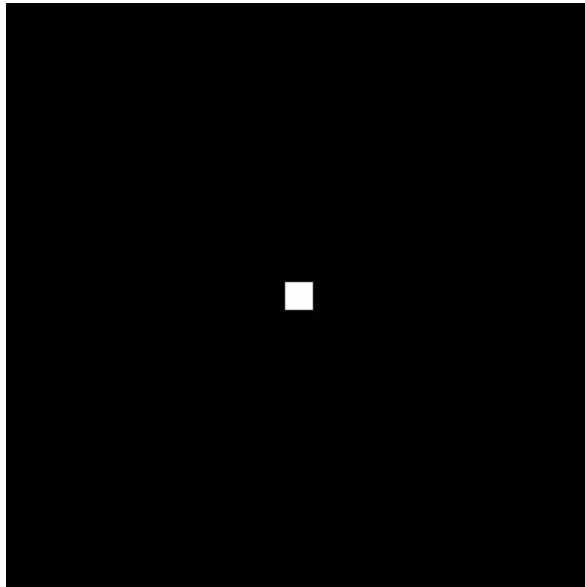
Sample

Note antialiased edges in rasterized triangle where pixel values take intermediate values

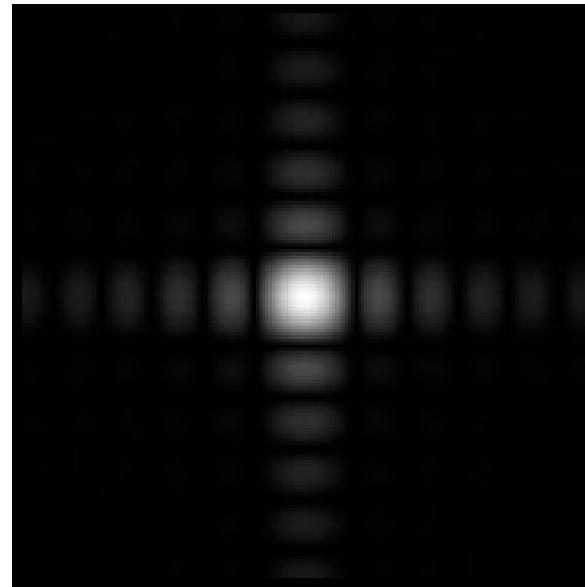


A Practical Pre-Filter

A 1 pixel-width box filter (low pass, blurring)

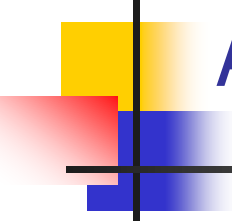


Spatial Domain



Frequency Domain





Antialiasing By Averaging Values in Pixel Area

Solution:

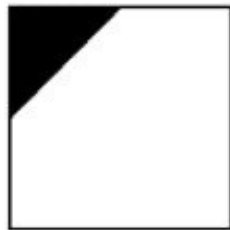
- **Convolve** $f(x,y)$ by a 1-pixel box-blur
 - Recall: convolving = filtering = averaging
- **Then sample** at every pixel's center



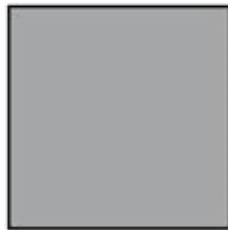
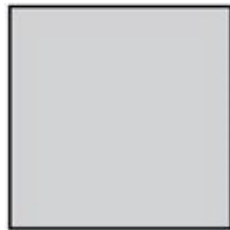
Antialiasing by Computing Average Pixel Value

In rasterizing one triangle, the average value inside a pixel area of $f(x,y) = \text{inside}(\text{triangle}, x, y)$ is equal to the area of the pixel covered by the triangle.

Original

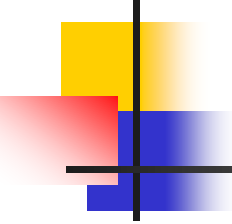


Filtered



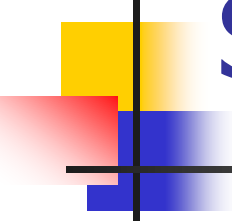
1 pixel width





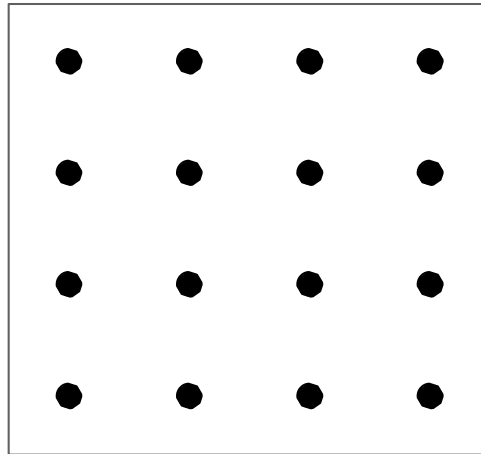
Antialiasing By Supersampling (MSAA)





Supersampling

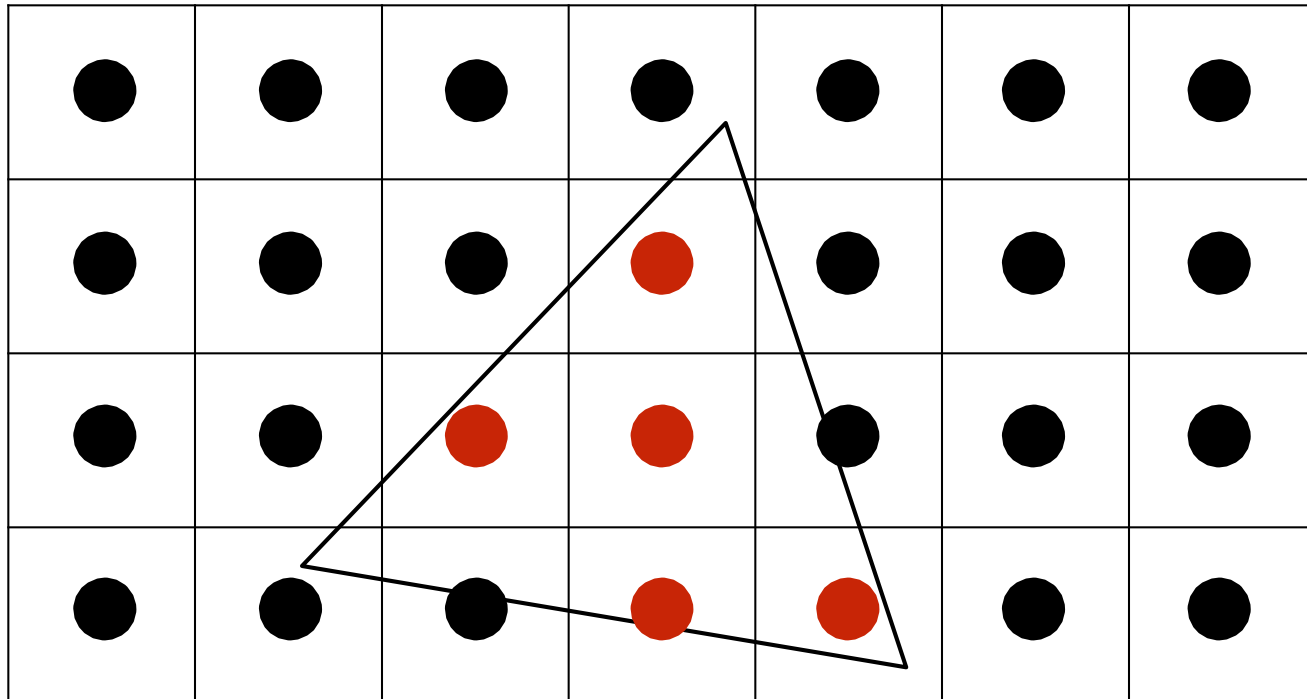
Approximate the effect of the 1-pixel box filter by sampling multiple locations within a pixel and averaging their values:



4x4 supersampling

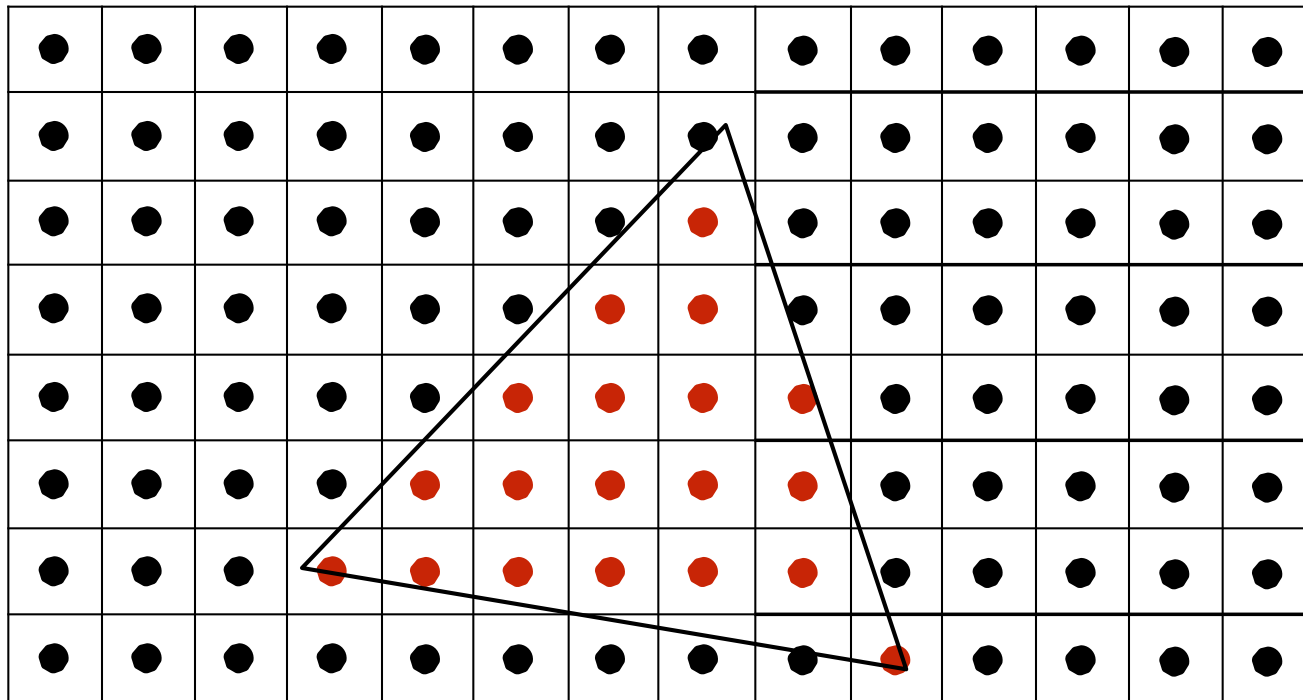


Point Sampling: One Sample Per Pixel



Supersampling: Step 1

Take $N \times N$ samples in each pixel.

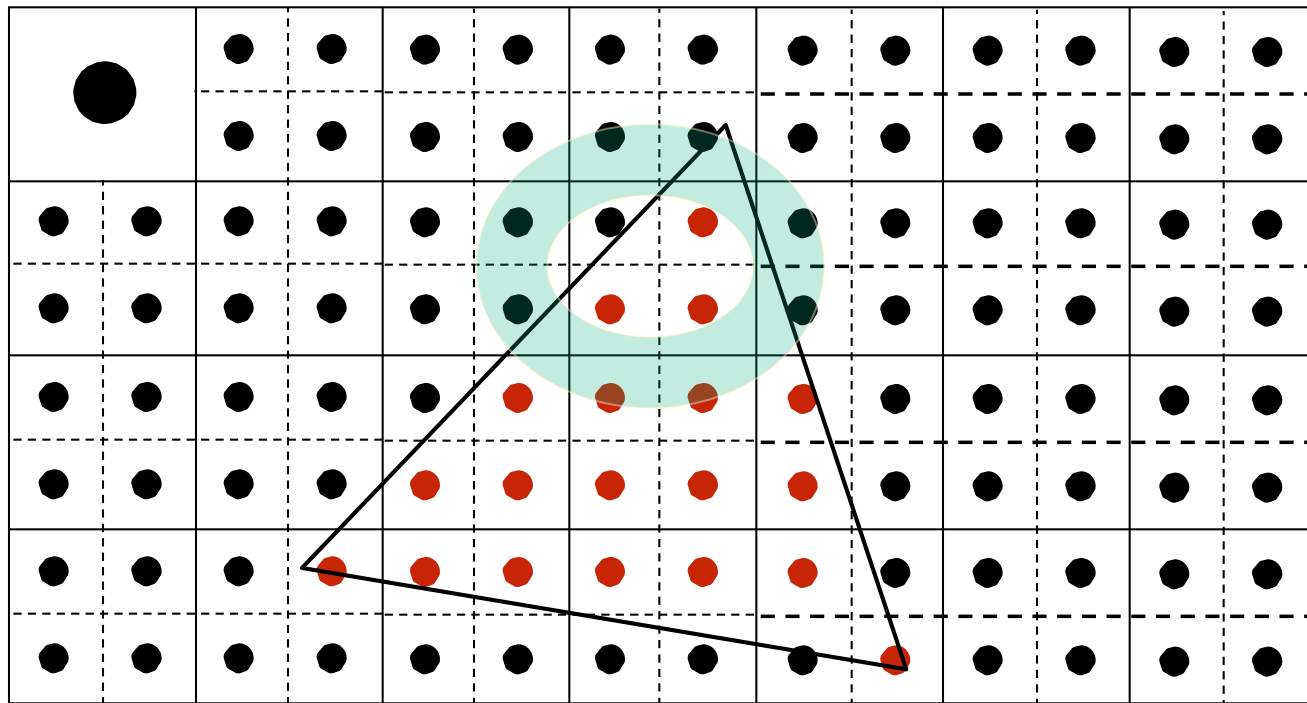


2x2 supersampling



Supersampling: Step 2

Average the $N \times N$ samples “inside” each pixel.

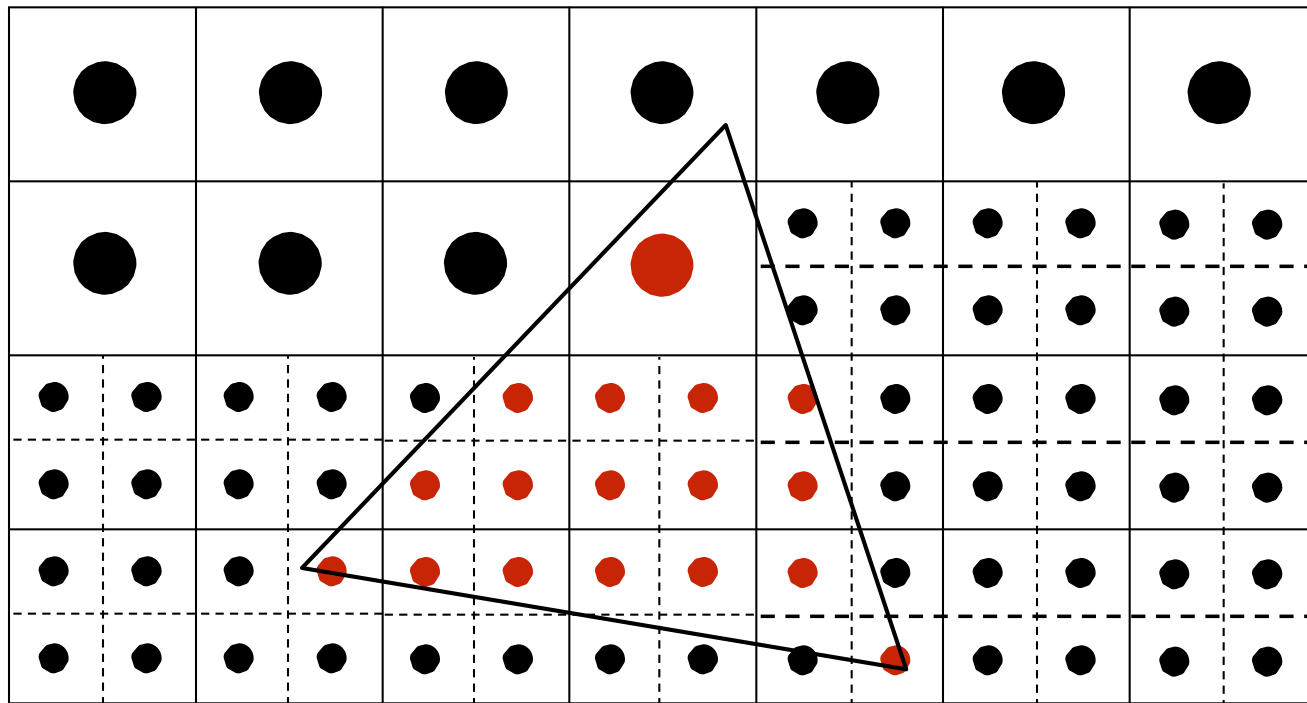


Averaging down



Supersampling: Step 2

Average the $N \times N$ samples “inside” each pixel.

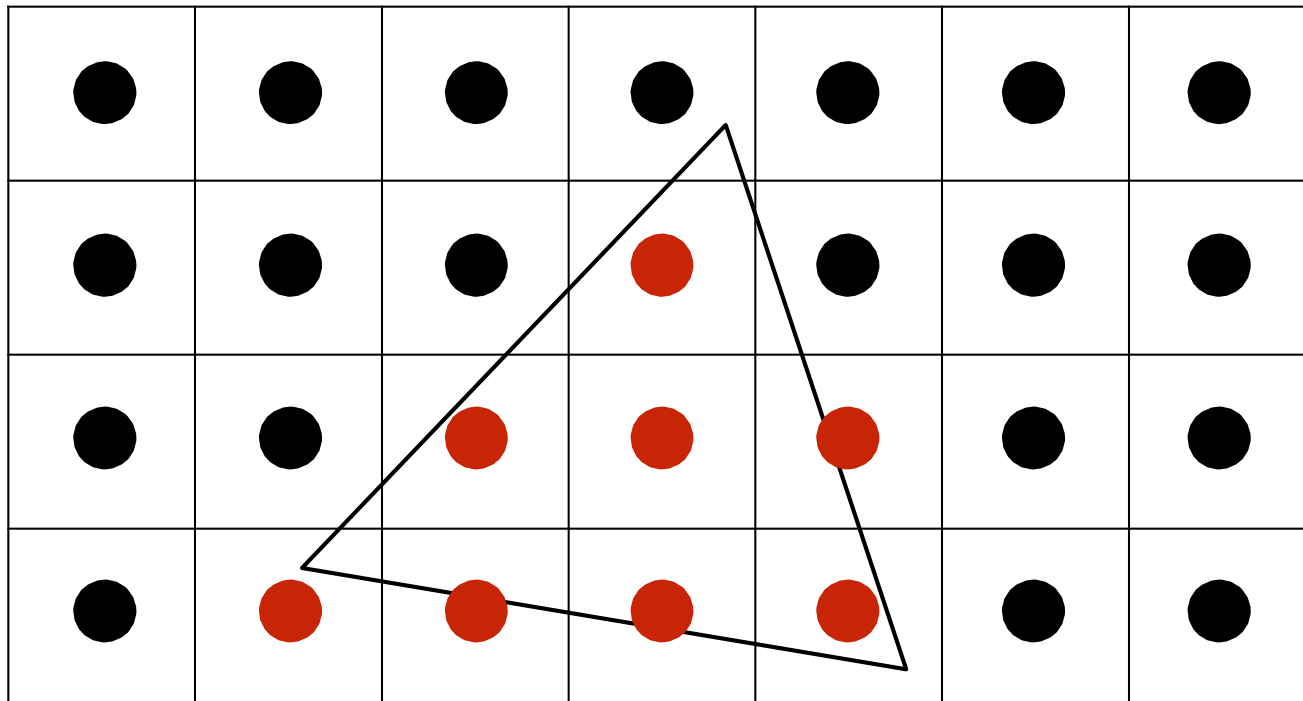


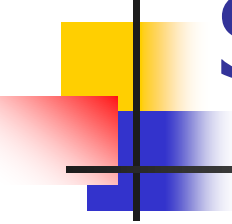
Averaging down



Supersampling: Step 2

Average the $N \times N$ samples “inside” each pixel.





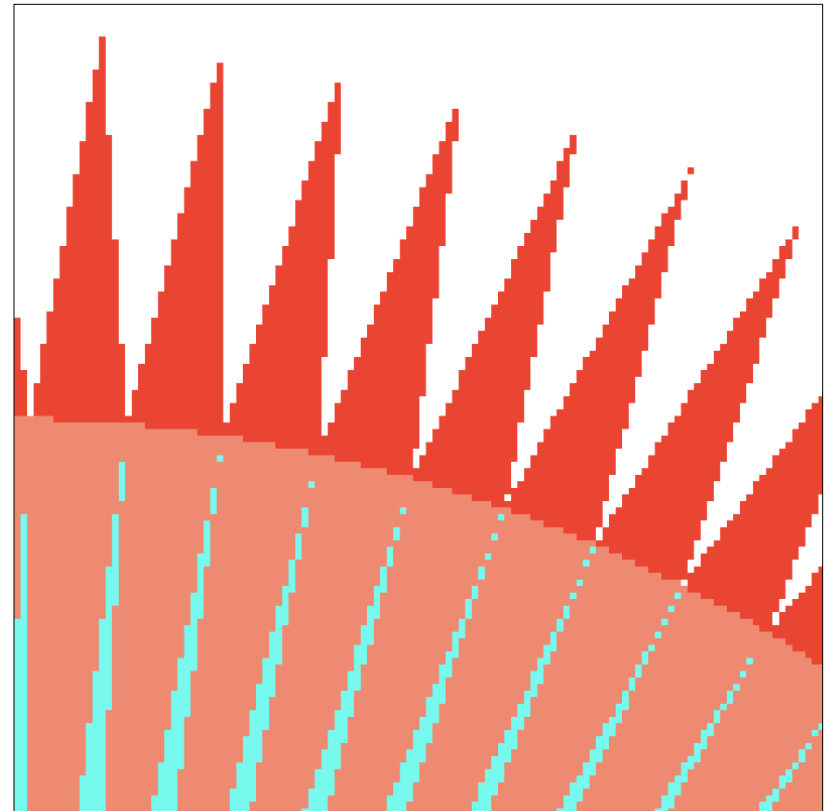
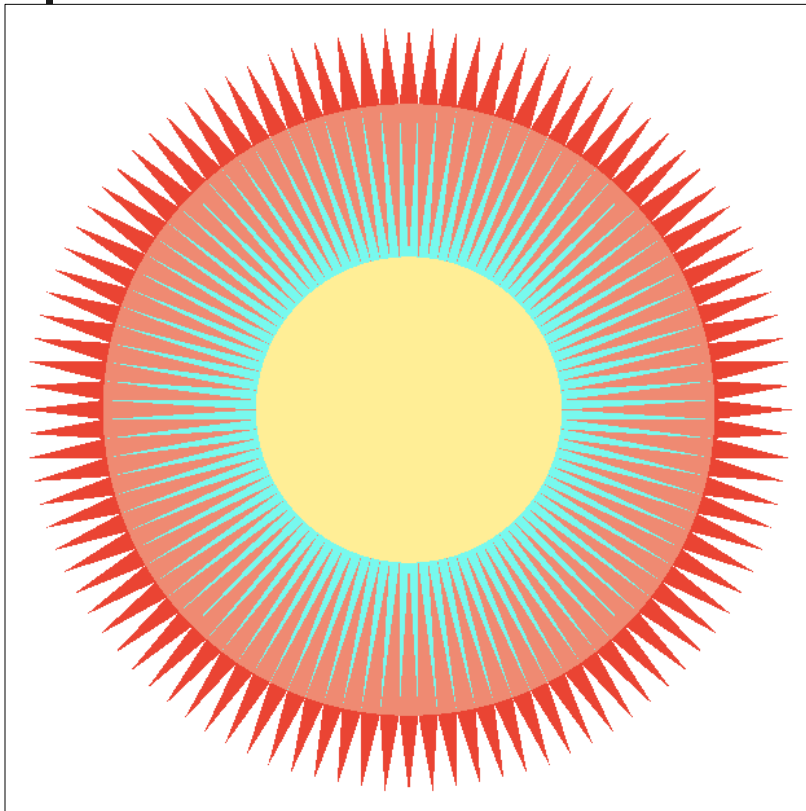
Supersampling: Result

This is the corresponding signal emitted by the display

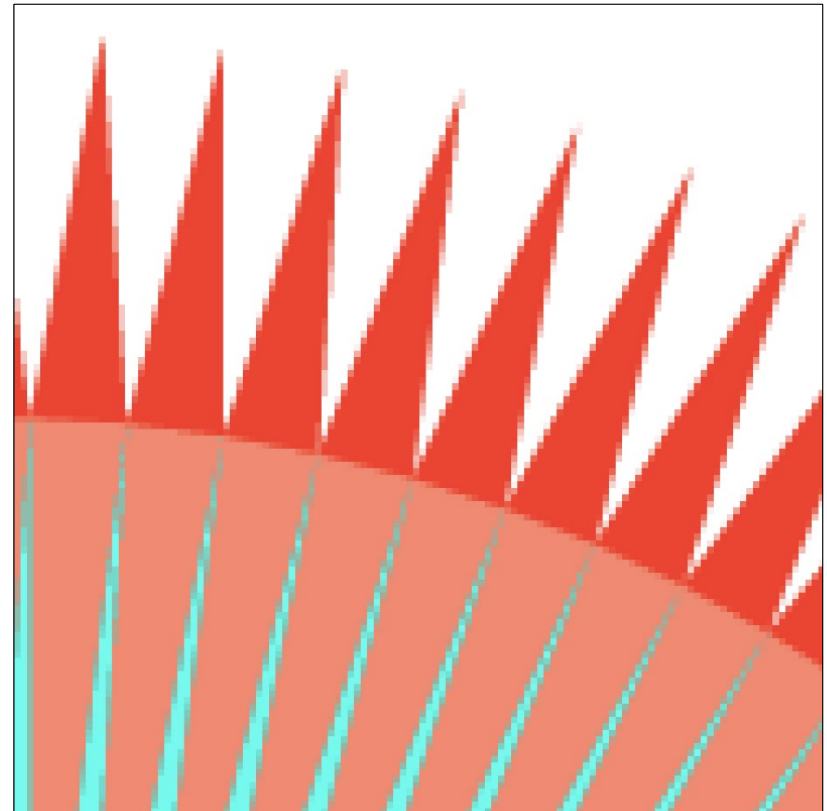
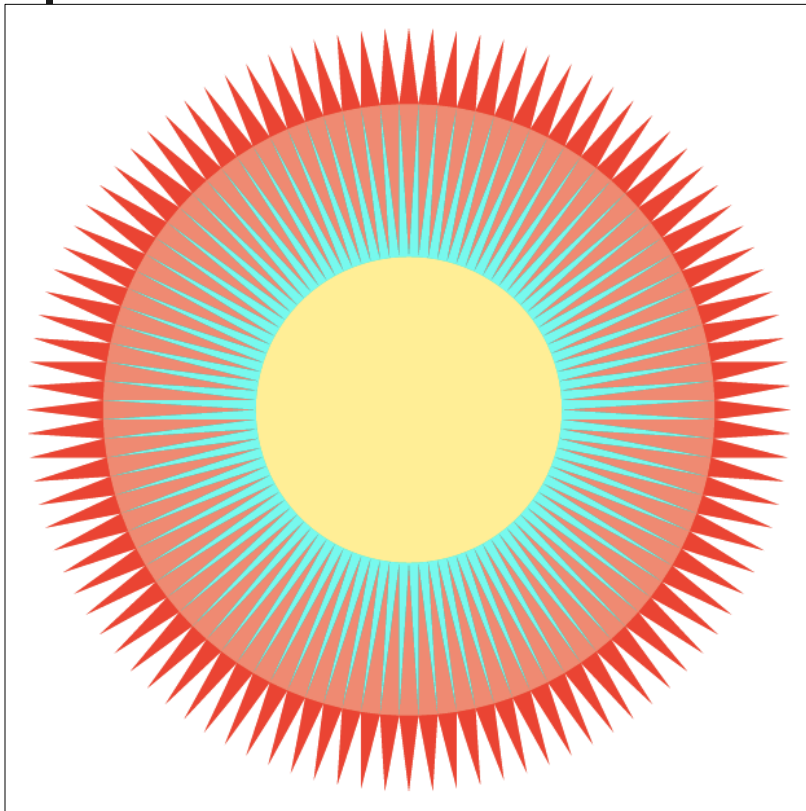
			75%			
		100%	100%	50%		
	25%	50%	50%	50%		



Point Sampling



4x4 Supersampling





Antialiasing Today

No free lunch!

- What's the cost of MSAA?

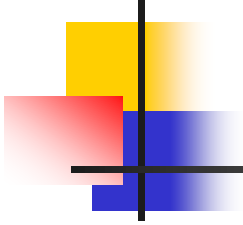
Milestones

- FXAA (Fast Approximate AA)
- TAA (Temporal AA)

Super resolution / super sampling

- From low resolution to high resolution
- Essentially still “not enough samples” problem
- DLSS (Deep Learning Super Sampling)





Thank you!

(And thank Prof. lingqi Yan (UCSB) for many of the slides!)



UNIVERSITY *of* **HOUSTON**

DEPARTMENT OF COMPUTER SCIENCE